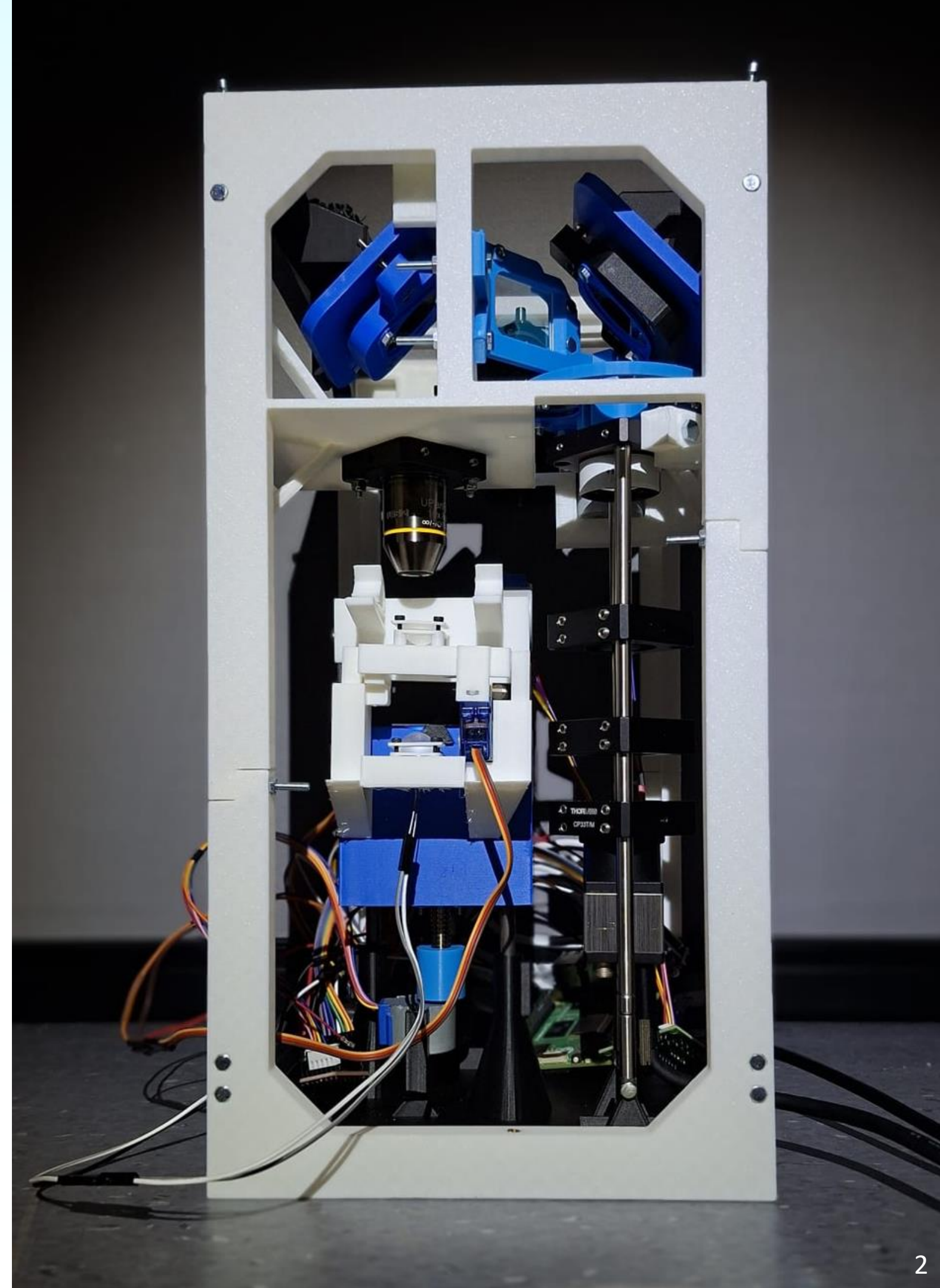


TURTLE

Transportable Universal Research
Tool for Live Examination

- Eliminates sample transfer
- Fluorescence, brightfield and darkfield
- Suitable for every manufacturer



General Information

PRICE (OPTICAL COMPONENTS EXCLUDED):

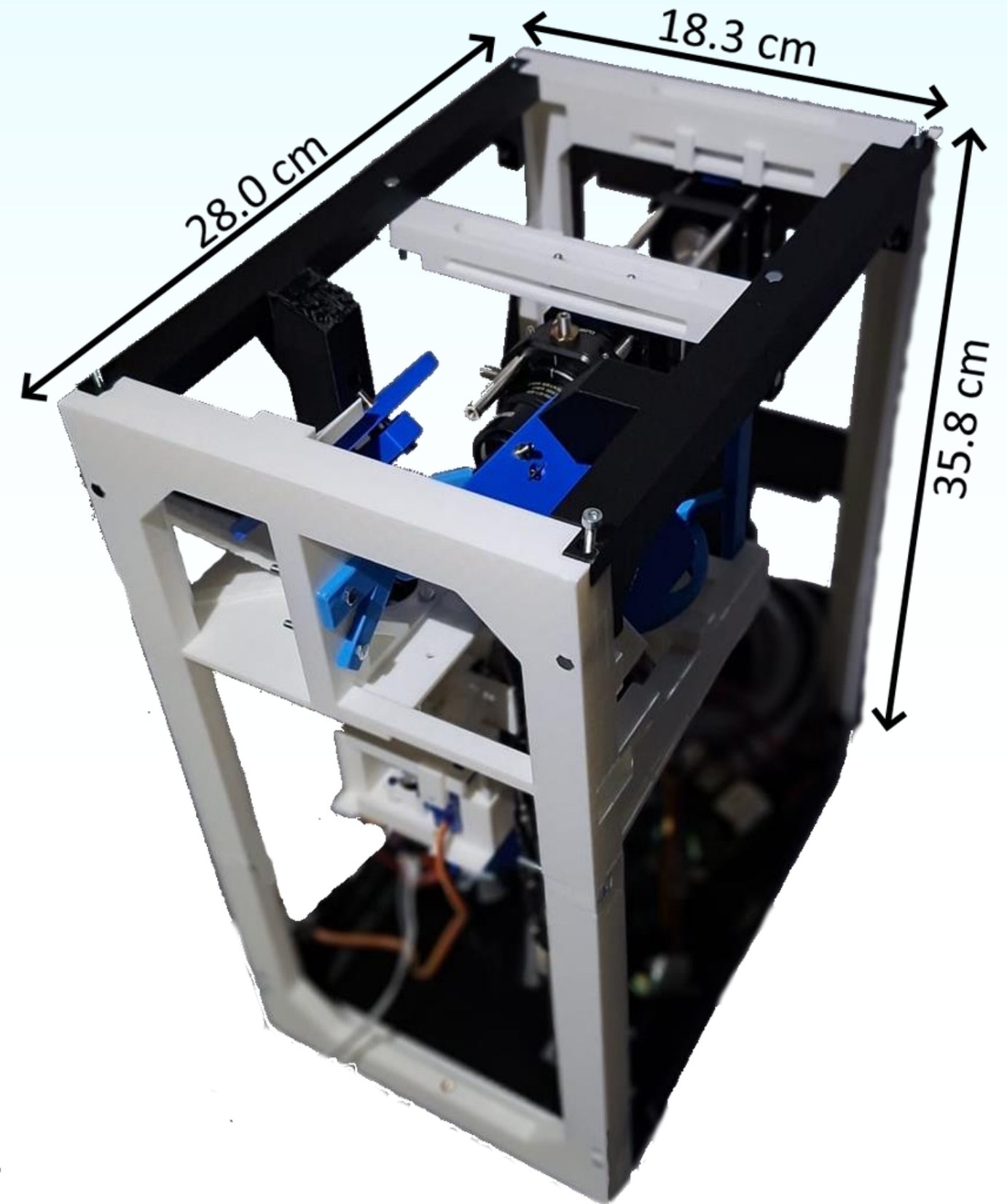
€779,46

WEIGHT:

2.2 kg

DIMENSIONS:

35.8 × 28.0 × 18.3 cm



Optical path

THEORETHICAL MAGNIFICATION:

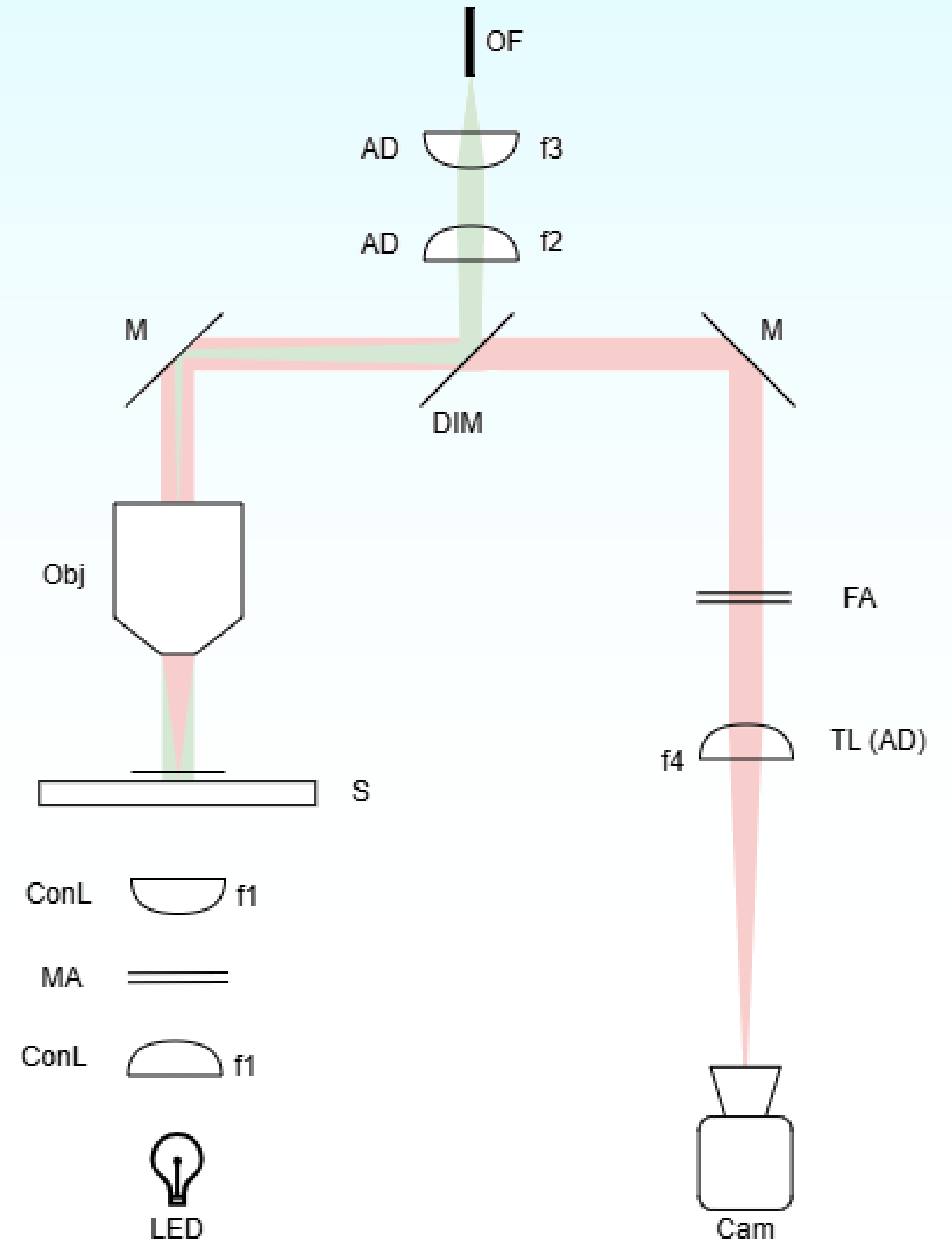
8.33x

RAYLEIGH'S CRITERION:

1.057 μm

CAMERA RESOLUTION:

0.414 μm



Legend: OF = optical fiber, M = mirror, DIM = dichroic mirror, Obj = objective lens, S = sample, ConL = condensor lens, MA = mask array, FA = filter array, TL = tube lens, AD = achromatic doublet, Cam = camera

f1 = 12 mm, f2 = 150 mm, f3 = 75 mm, f4 = 150 mm

Optical path

THEORETHICAL MAGNIFICATION:

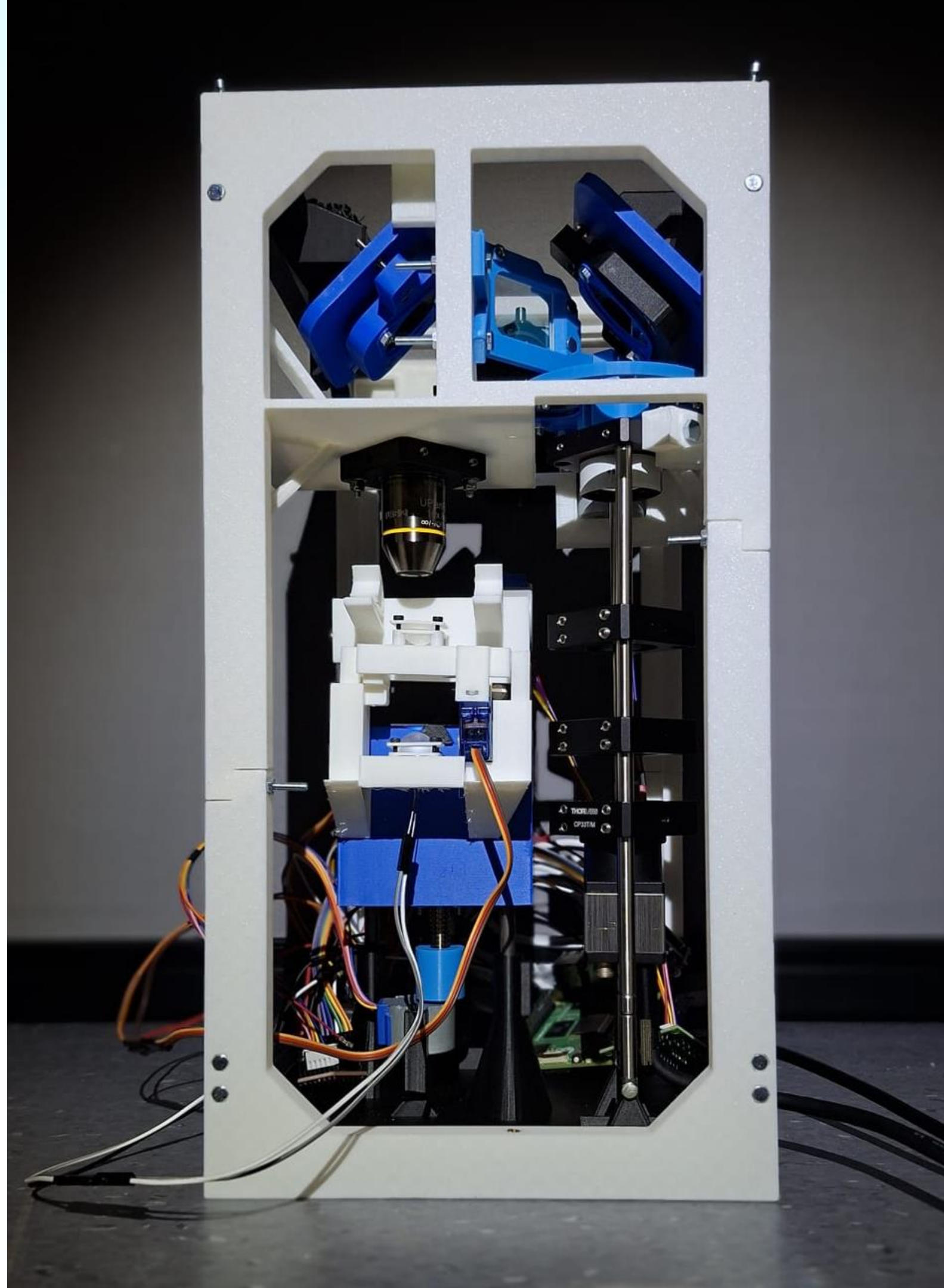
8.33x

RAYLEIGH'S CRITERION:

1.057 μm

CAMERA RESOLUTION:

0.414 μm



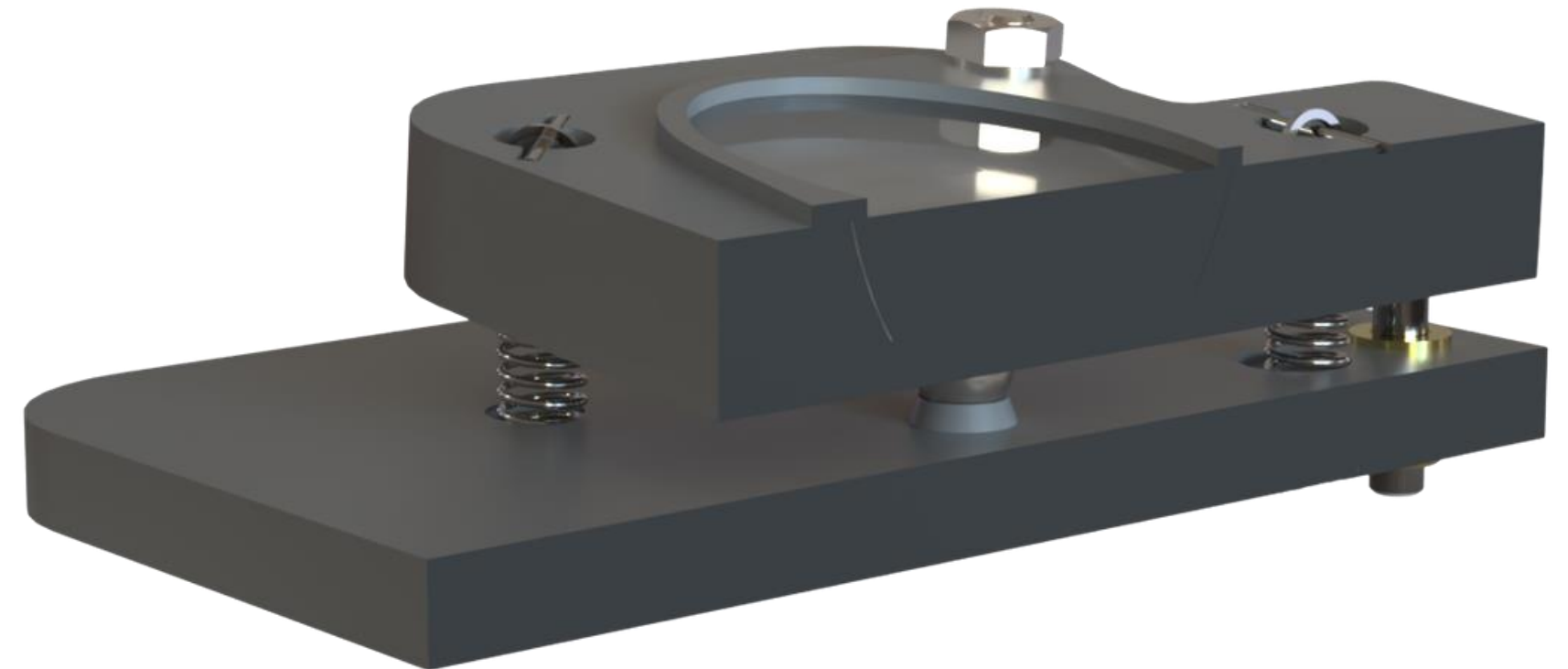
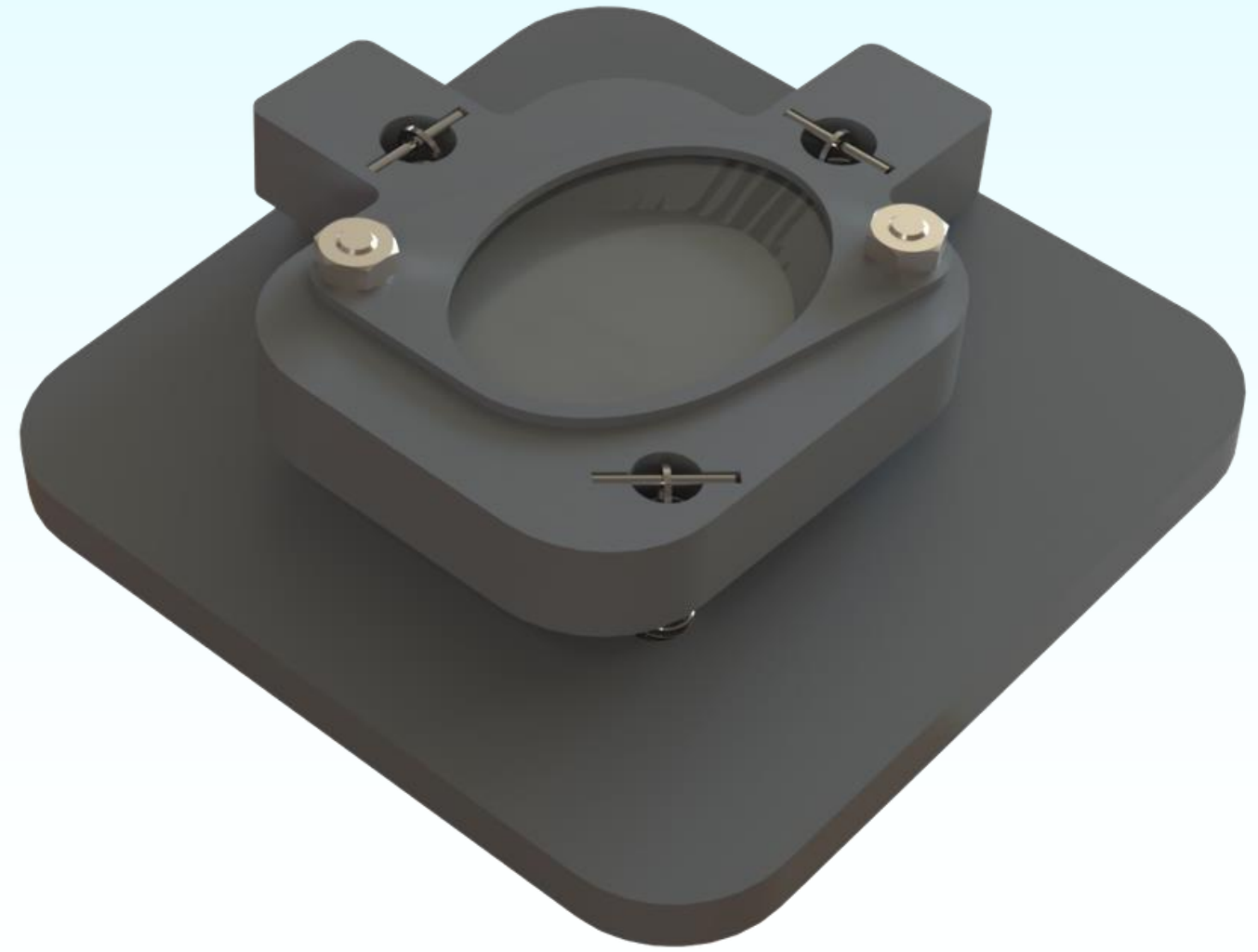
Mirror Mount

ADJUSTEMENT RESOLUTION:

$0.37^\circ/\text{rev}$

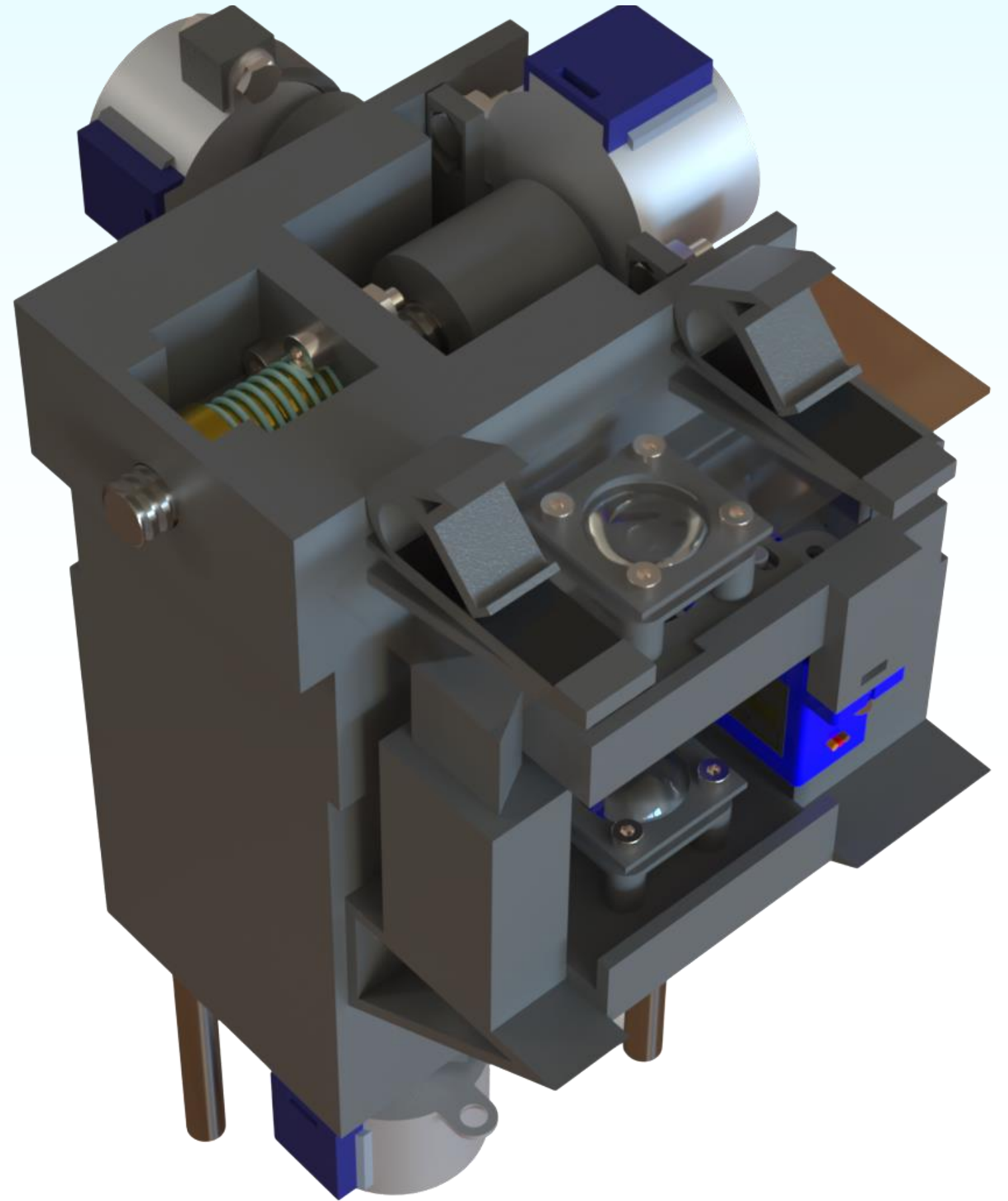
RANGE:

$\pm 5^\circ$

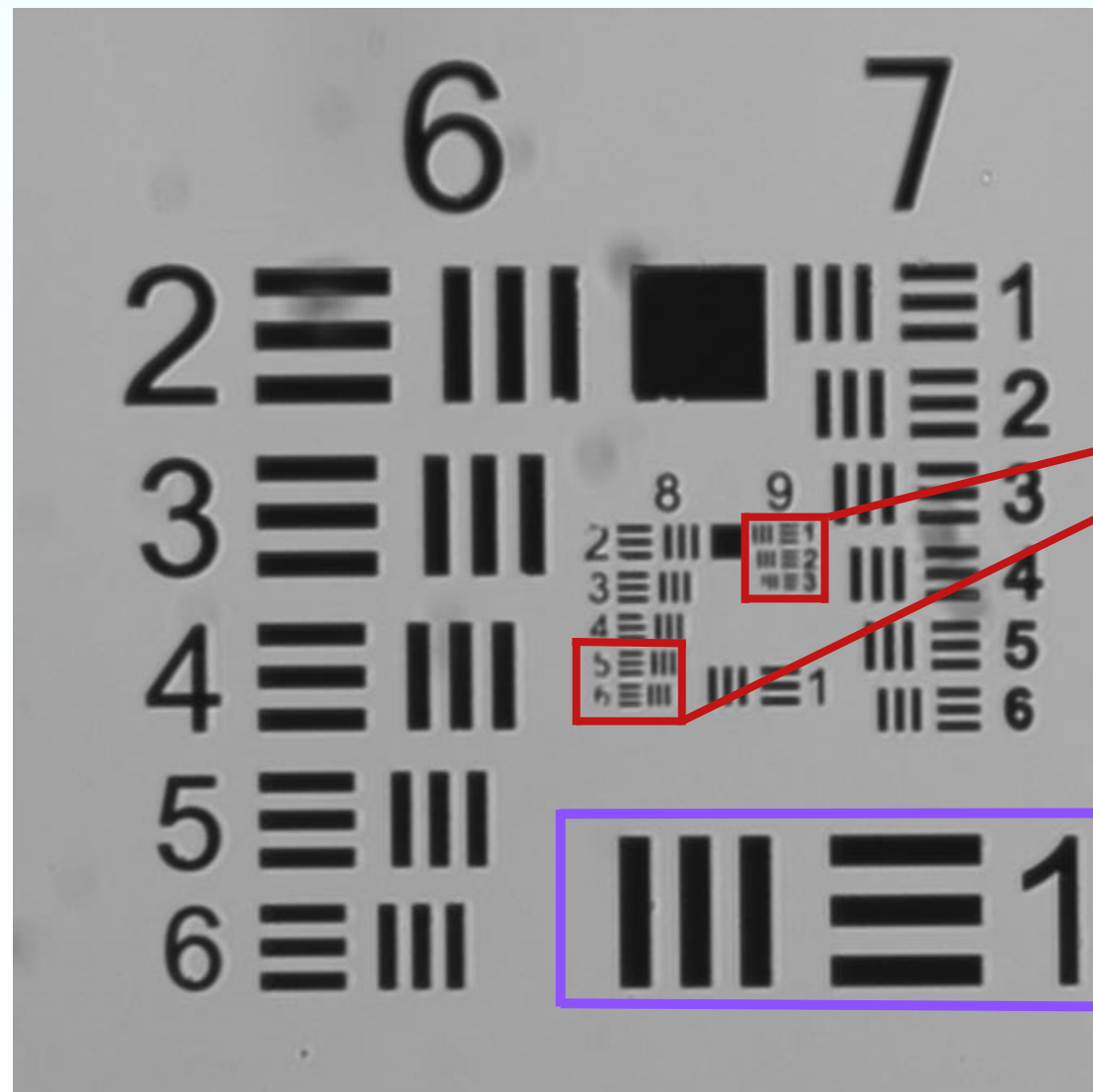


Stage

- Adjustment accuracy:
0.48 $\mu\text{m}/\text{step}$
- Range of:
 - Y: 10mm
 - X: 8mm
 - Z: 35mm
- Vibrations



USAF PS75P



Resolution

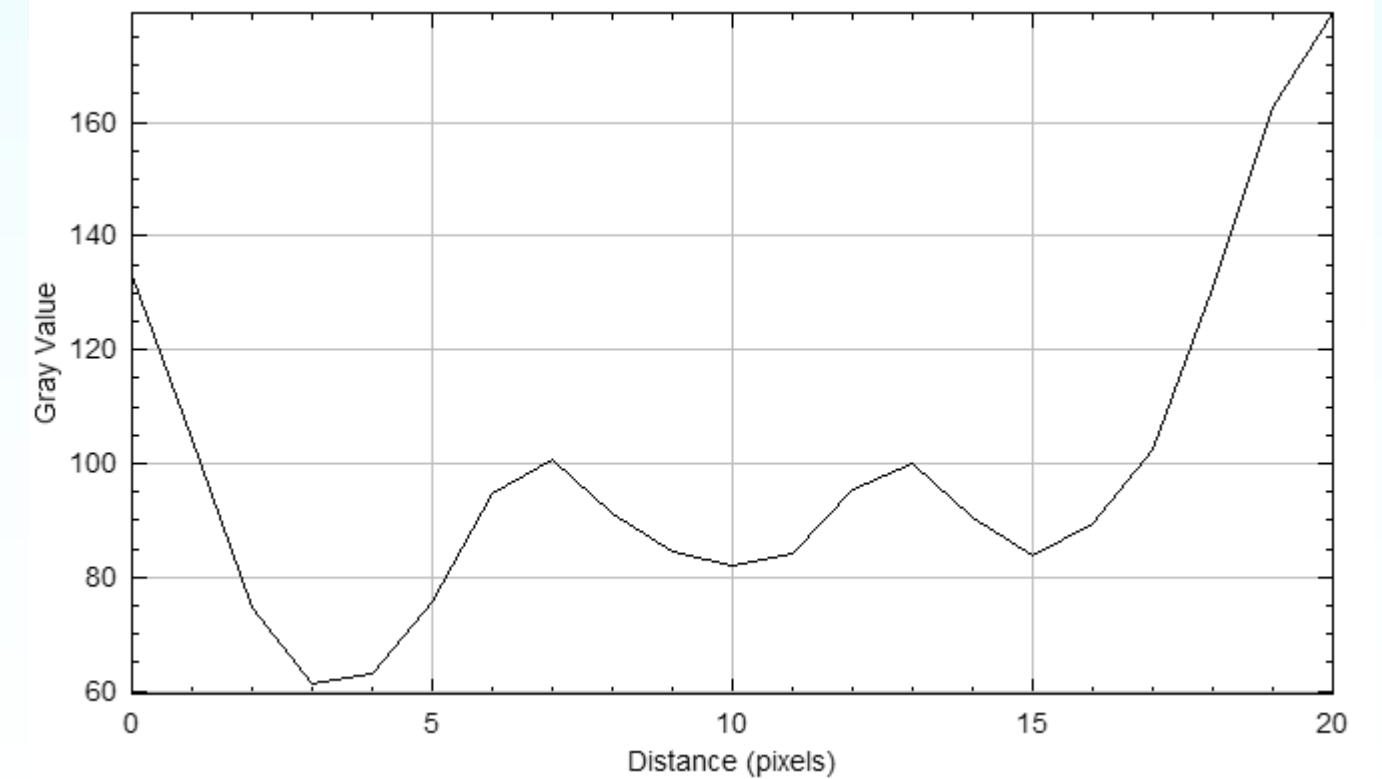
Estimated
Magnification



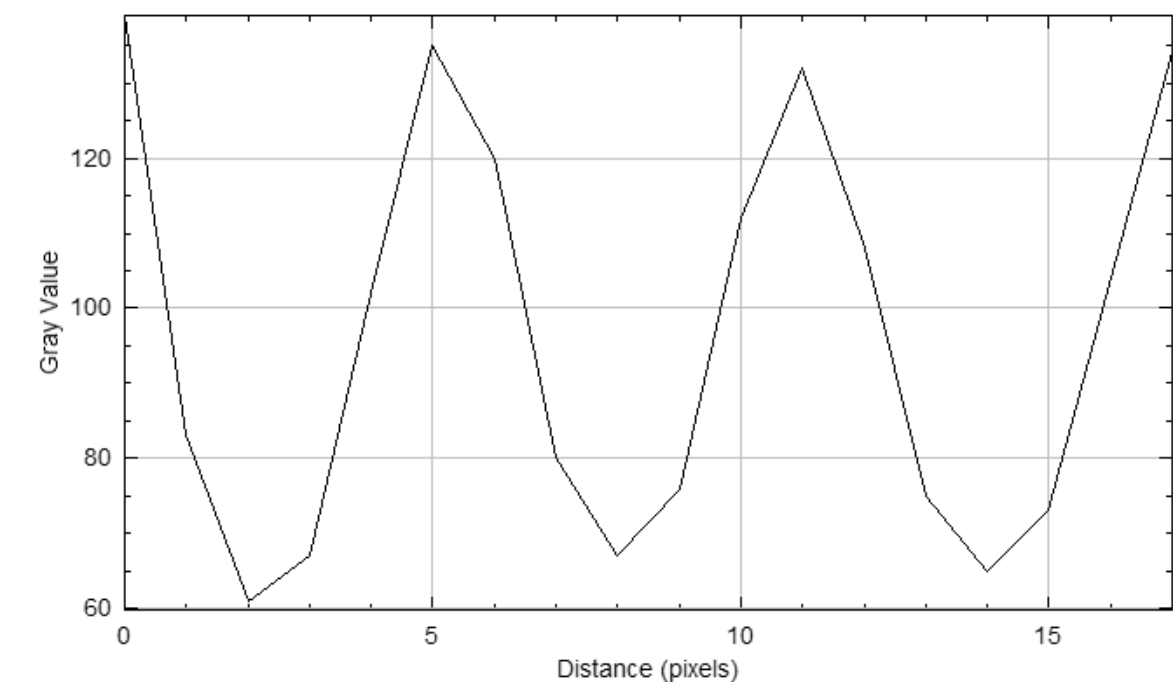
USAF Resolution

- Theoretical values: 1.057 μm
- 5 Samples
- Required dip of 26.3%
- Analysed interspacings: 1.2 μm to 0.78 μm

Result:
1.2 μm



The intensity curve of group 8 element 5 of sample 4 (measured vertically)



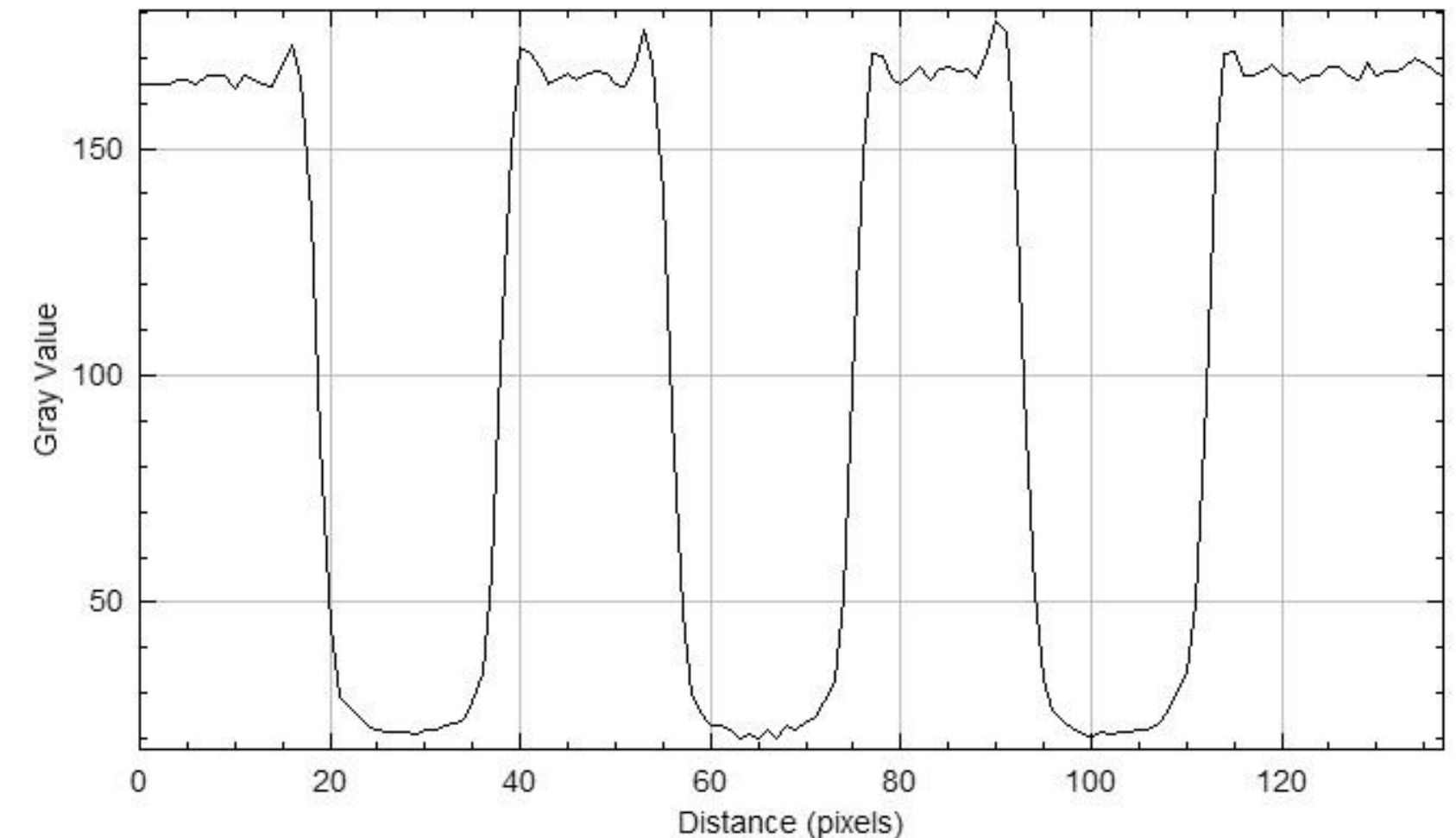
The intensity curve of group 8 element 5 of sample 4 (measured horizontally)

Estimated Magnification

- Theoretical values: 8.33
- 5 Samples
- USAF PS75P: Group 6, element 1
- Full width at half maximum

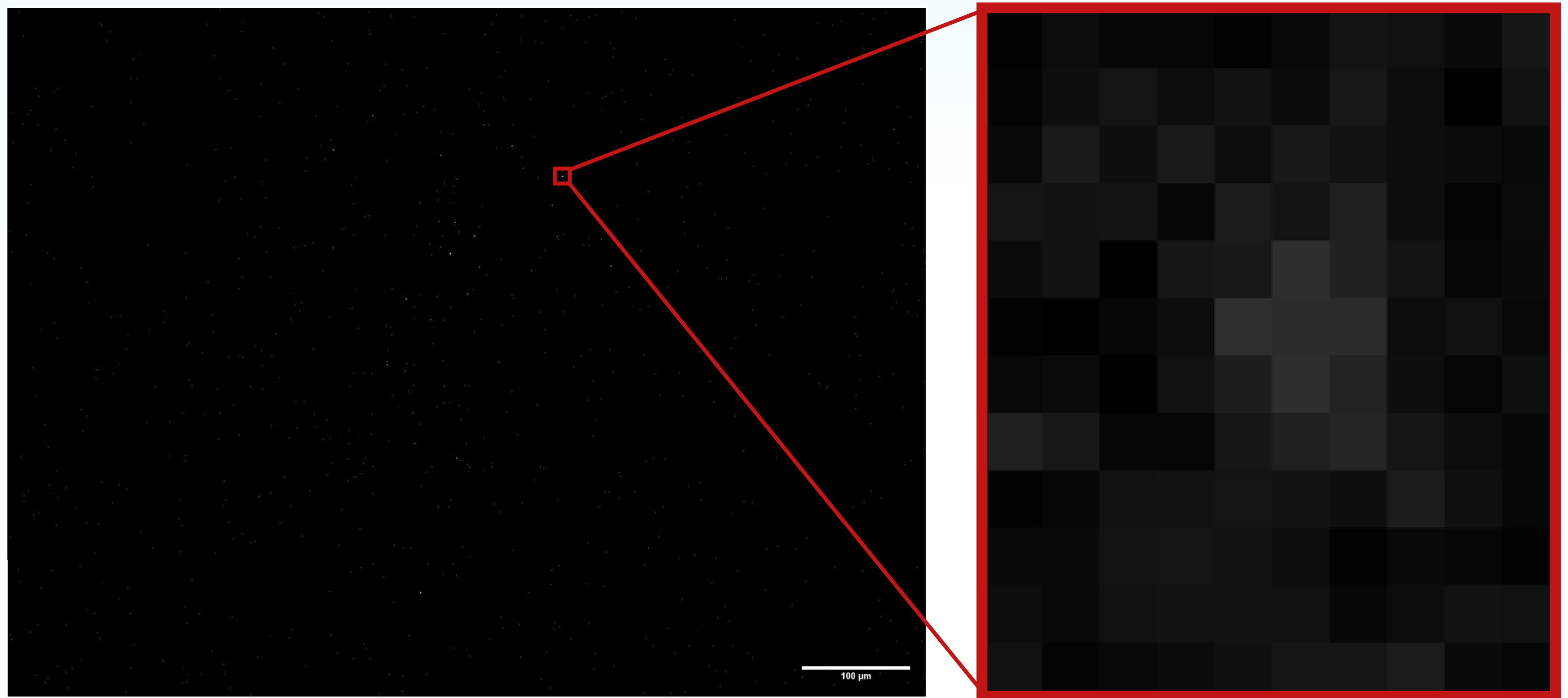
Result:

8.45 ± 0.08

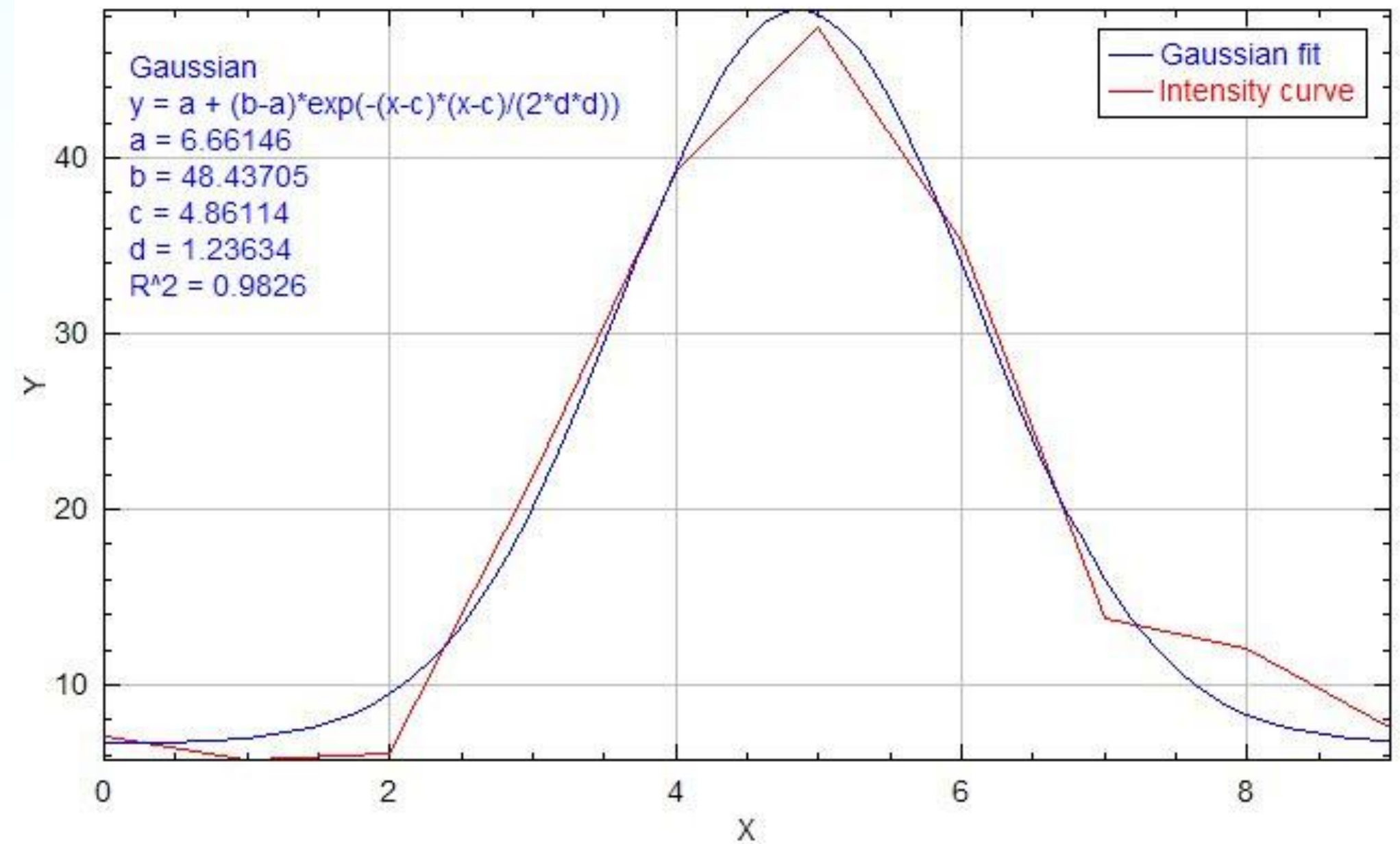
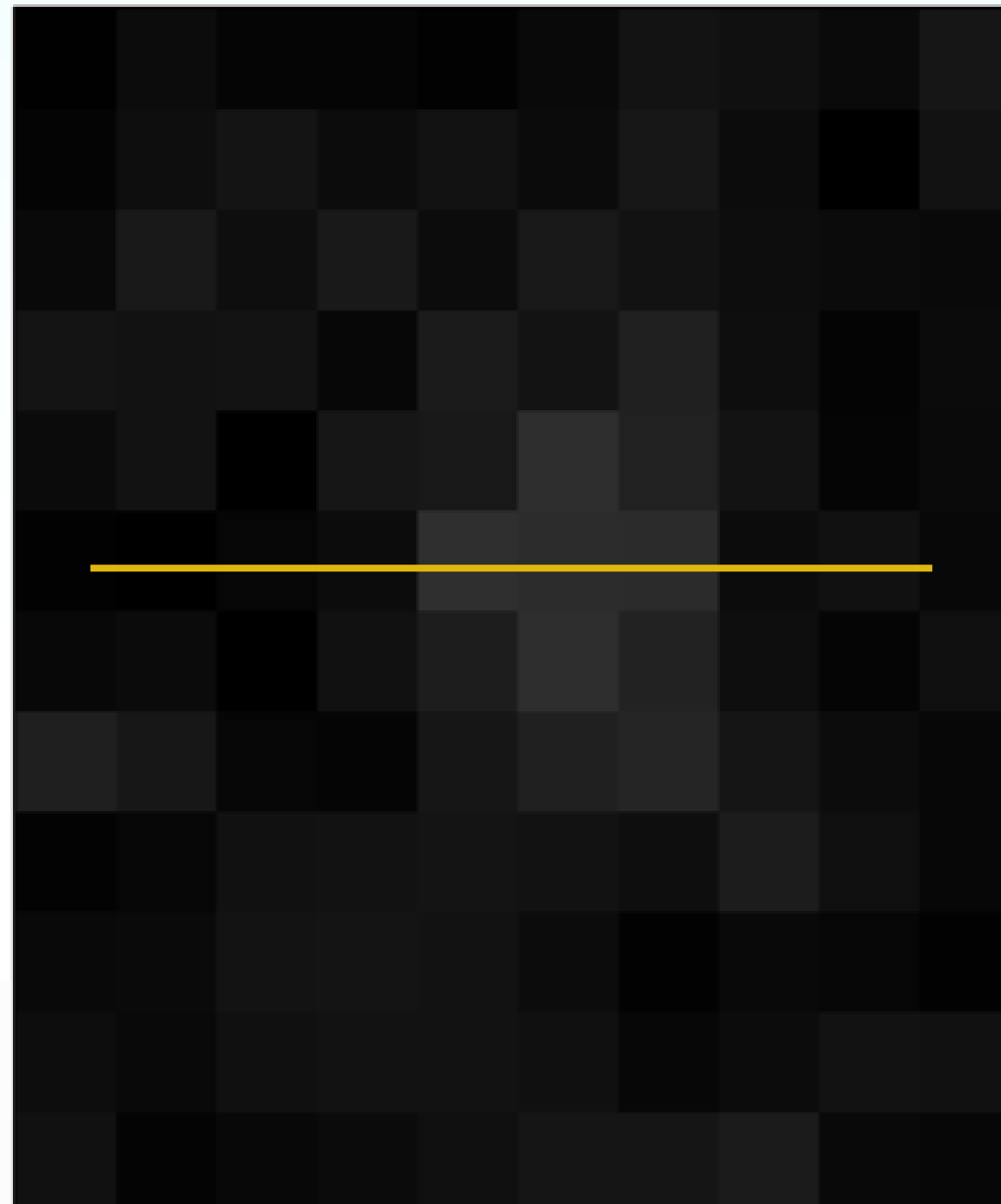


The intensity curve of group 6 element 1 of sample 1 (measured horizontally)

Point Spread Function



Point Spread Function



Point Spread Function

SIMULATED FWHM:

1 μm

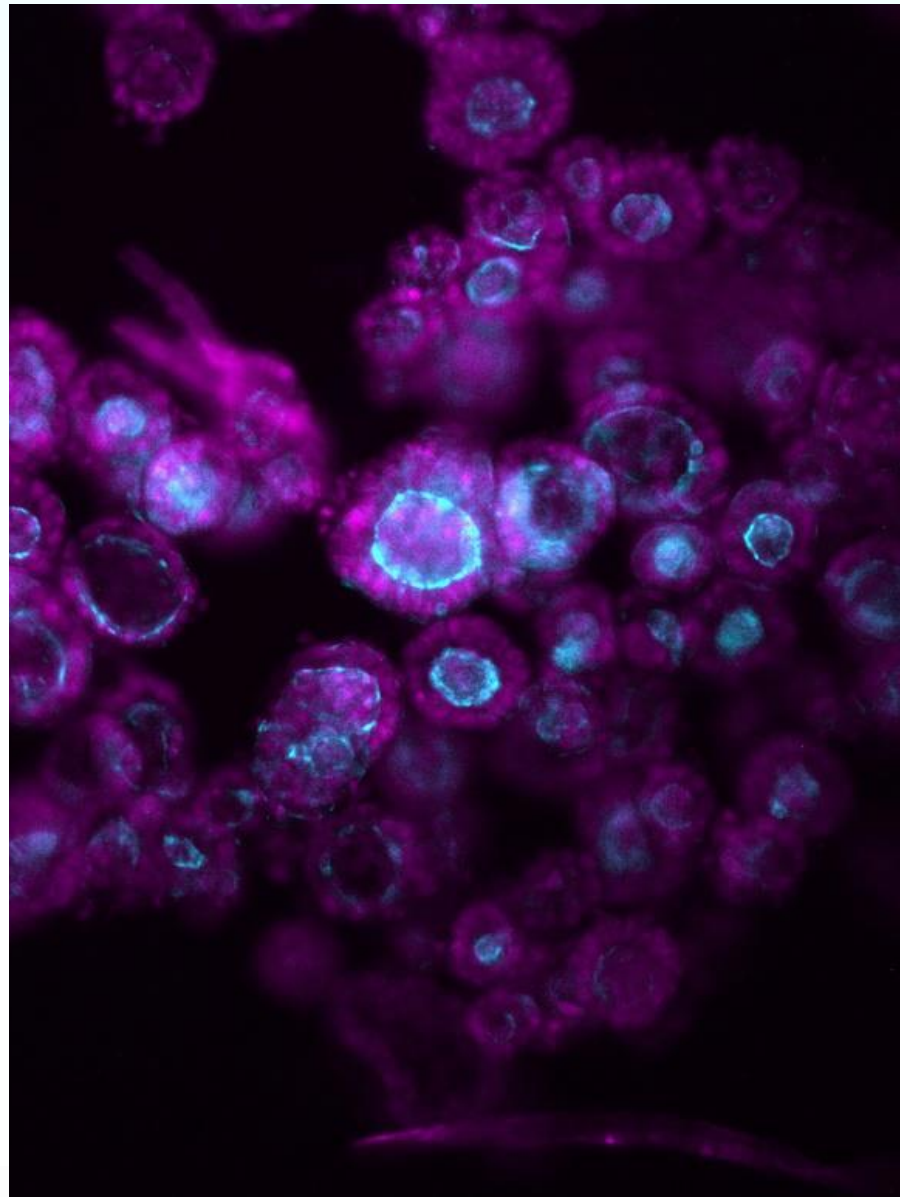
MEASURED FWHM:

1.39 $\pm$ 0.01 μm

Bead	FWHM in x (μm)	FWHM in y (μm)
1	1.18306	1.50871
2	1.41671	1.19935
3	1.43033	1.42807
4	1.58994	1.88634
5	1.38138	1.29675
6	1.51651	1.24439
7	1.63190	1.15266
8	1.25915	1.32867
9	1.06627	1.56145
10	1.10833	1.58044
Average	1.36	1.42
Standard deviation	0.19748	0.22253
Coefficient of Validation (%)	14.53807	15.68552

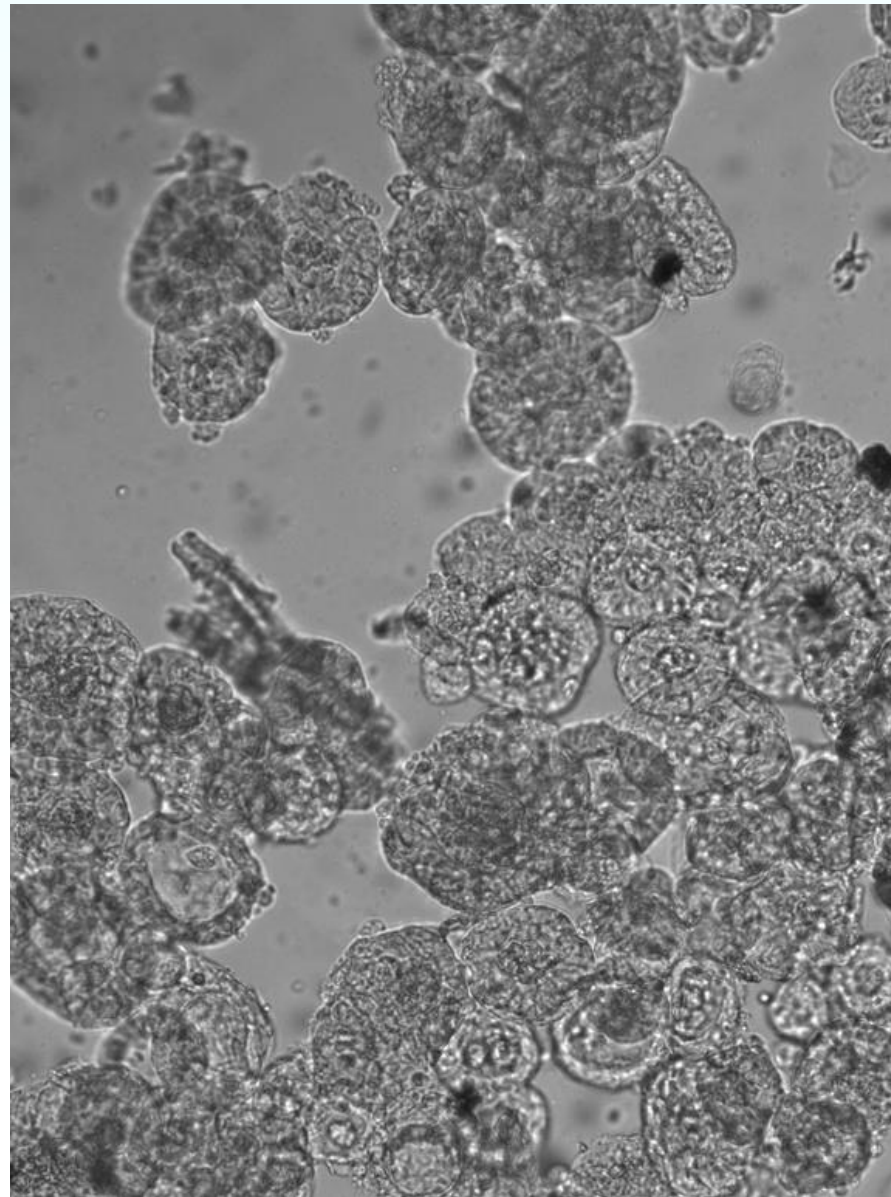


Images



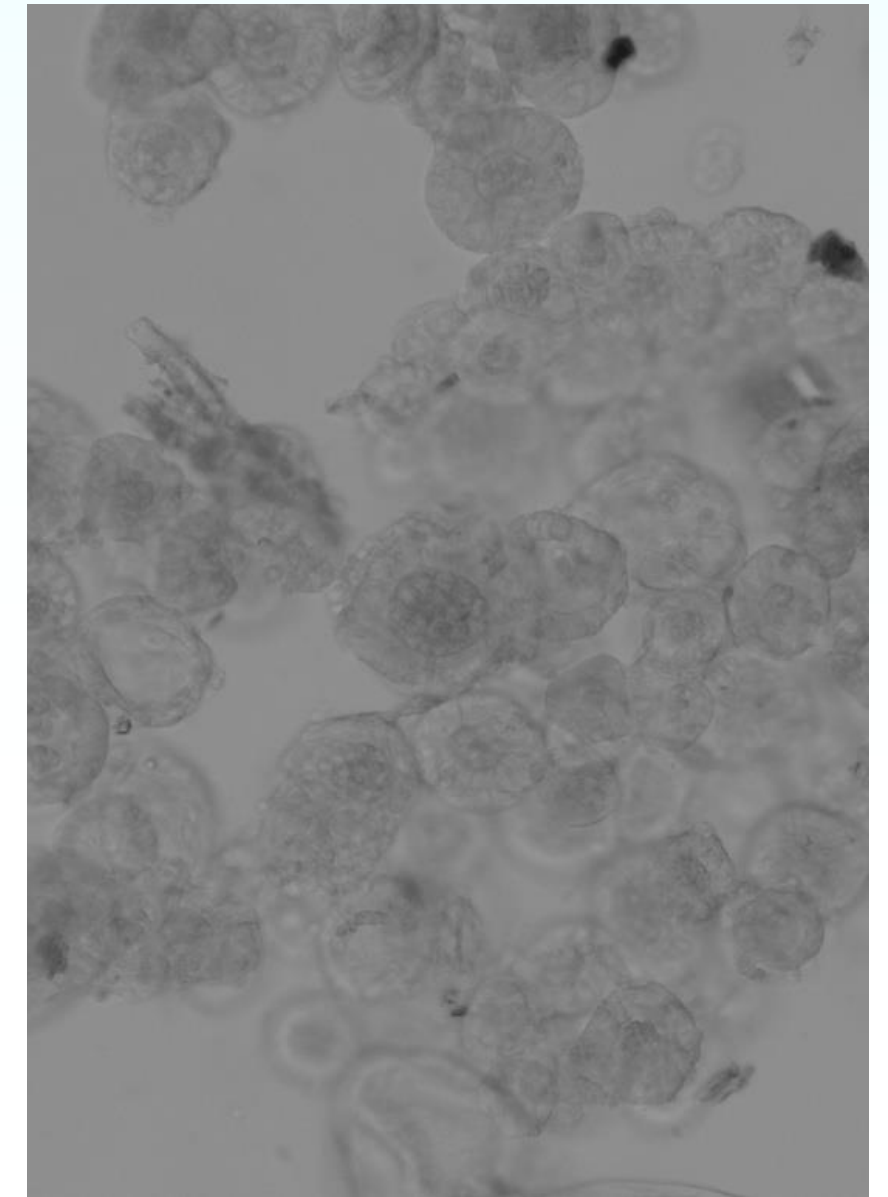
Fluorescence

Sample supplied by
Federico Berto



Brightfield

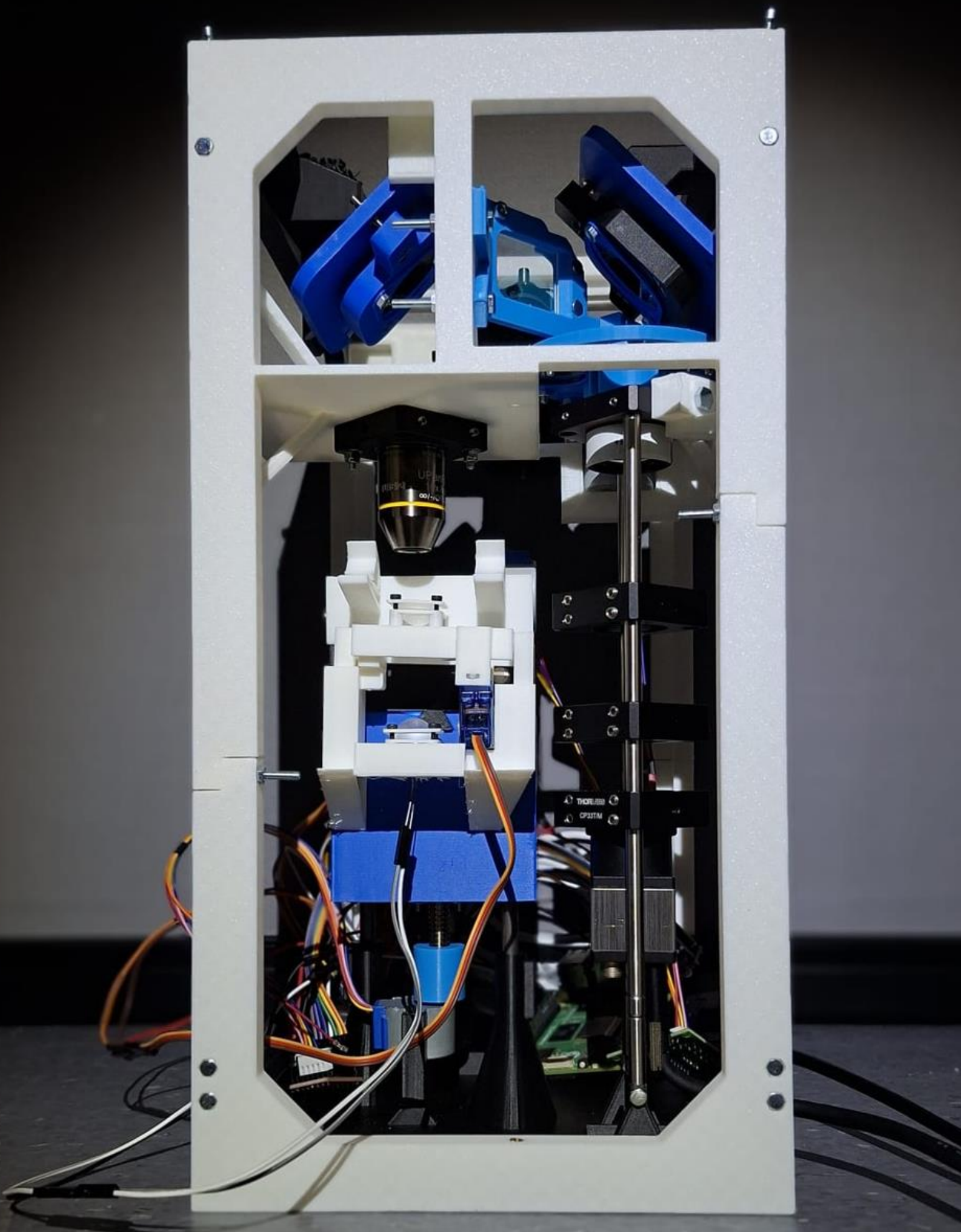
Sample supplied by
Federico Berto



Darkfield

Sample supplied by
Federico Berto





TURTLE

To eliminate sample transfer



Fluorescence, widefield and darkfield

Mirror accuracy:

0.37 °/rev

Adjustment accuracy:

0.48 μm /step

Determined resolution:

1.2 μm

Estimated magnification:

8.45x

Actual FWHM:

1.39 μm

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- [BALL JOINT BILL OF MATERIAL](#)
- [GIMBAL BILL OF MATERIAL](#)
- [1" MIRRORS COMPARISON](#)
- [ELLIPTICAL MIRRORS COMPARISON](#)
- [Stage](#)
- [MIRROR DRIFT TEST](#)
- [LASER MODULE](#)
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- [PSF Measurements](#)
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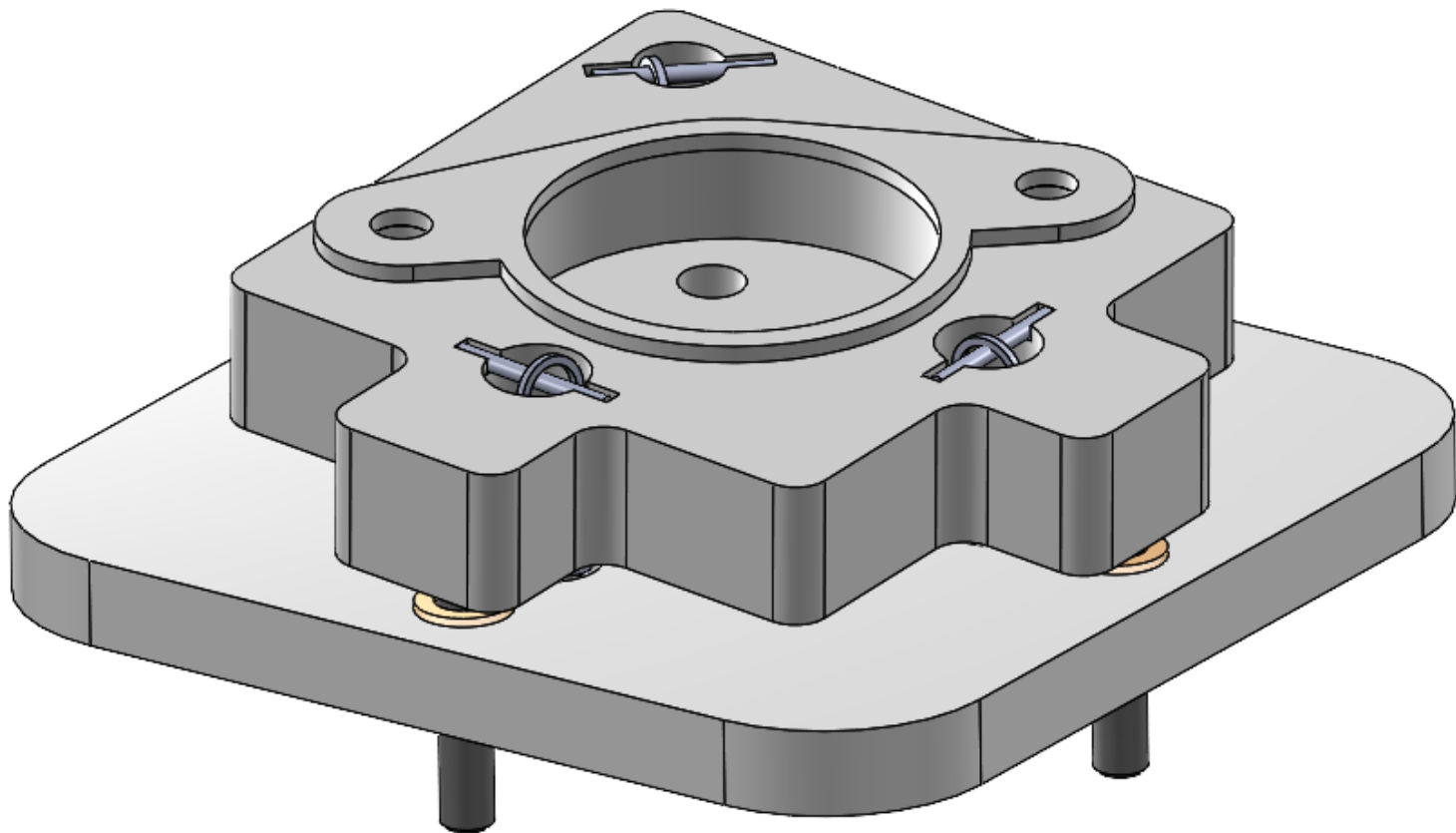
TURTLE Bill of Material

Component	Name	Amount	Price (€)	Component	Name	Amount	Price (€)
Objective lens	UPLFLN10X2	1	1200	Precision screws	F3ES15	4	20,08
Dielectric Elliptical Mirror	BBE1-E02	2	253,66	Precision screws bushings	F3ESN1P	4	17,52
Dichroic mirror	405/488/561/635 Quadband	1	714,47	Raspberry Pi 5	-	1	330,00
Emission filter	684/40 Avant™ Bandpass Filter	1	345,42	Raspberry Pico 2	-	1	5,00
Emission filter	MF525-39	1	306,38	Stepper motor driver (bipolar)	TB6612FNG	3	30,00
Achromatic Doublet, f=150mm	AC254-150-A-ML	1	113,23	Stepper motor driver (unipolar)	ULN2003	1	0,00
Camera	HIKrobot MV-CS060-10UM	1	431,77	Lead screws	TR8x2, 8 mm	1	9,50
Condenser lens, f = 12mm	ACL1512U-A	2	61,8	Steel rods	6x106 mm	2	-
LED	LED-5-18000W	1	0,38	Steel rods	4x24 mm	1	-
Stepper motor	28BYJ-48	4	48	Servo motor	SG90 Micro Servo	1	3,25
Beam shaping lenses f =10mm	AC080-010-A	2	106,92	Steel balls - 5mm	Technische kogels RVS 420	2	0,40
Cage plate	CP08/M	8	179,28	Filter wheel pin	3x8 mm	1	-
Cage rods 8"	ER8	2	24,62	Steel pin	1x10 mm	12	2,40
Cage rods 3"	ER3	4	27,76	Backlash nuts	anti-backlash, TR8x2	3	37,50
Cage rods 1/4"	ER025	2	10,62	Breadboard	400 pins	1	2,25
Cage rods extension	ER90C	2	27,50	Filament	PLA-filament	≈ 1 kg	22,50
Springs	T30876	2	6,74	Total:			4351,23
Springs	T31200	4	12,28				





Ball Joint Bill of Material

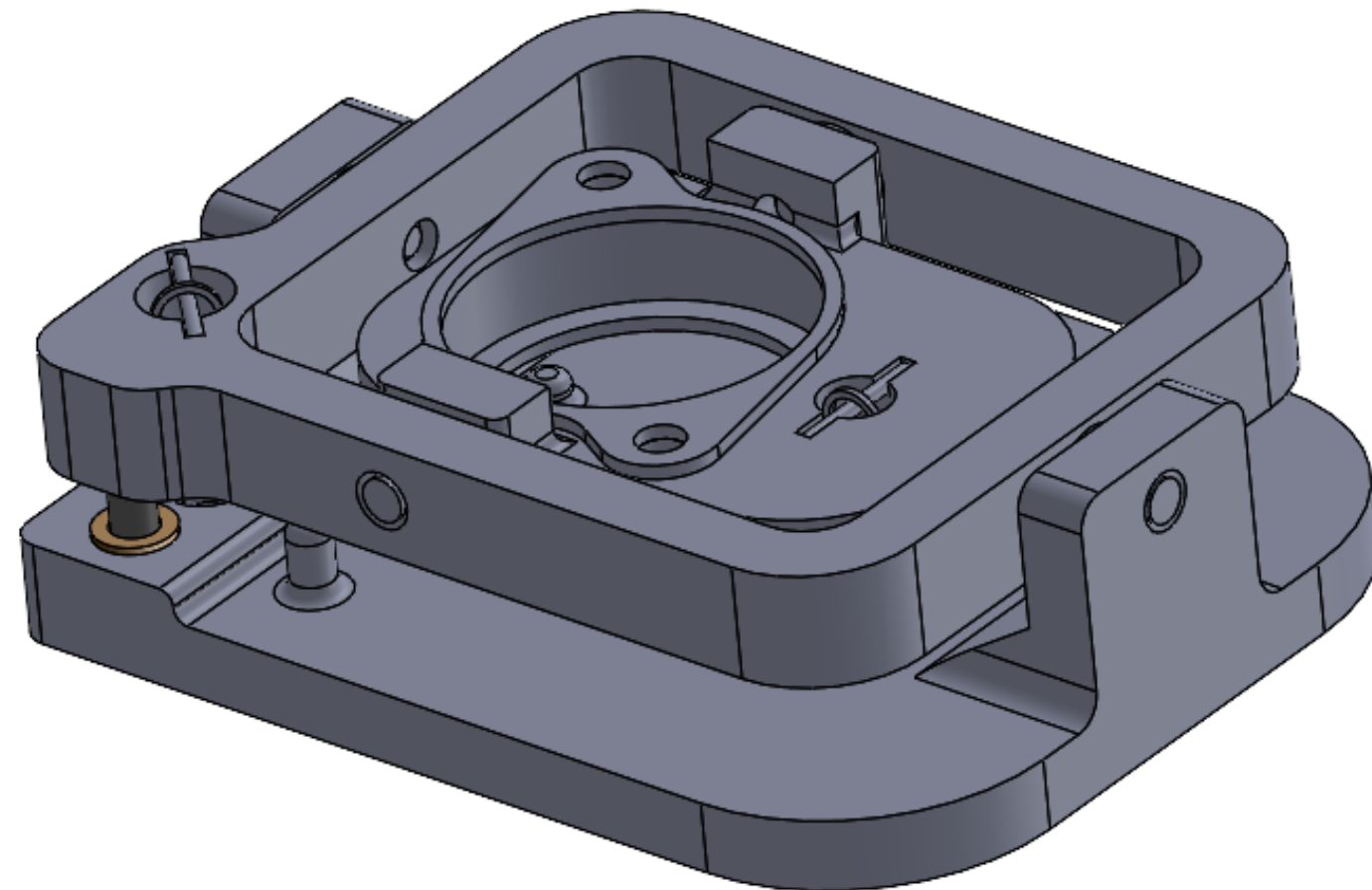


Tabel 1.1: Bill of materials per ball-jointed mirror mount

Component	Name	Amount
Backplate		1
Frontplate		1
Gasket		1
Springs	T31200	2
Springs	T30876	1
Precision screws	F3ESN1	2
Precision screws bushing	F3ESN1P	2
Steel ball	Technische kogels RVS 420	1
Steel pins	1x10mm	6
Elliptical/circular mirror	BBE1-E02	1



Gimbal Bill of Material



Tabel 1.2: Bill of materials per gimble mirror mount

Component	Name	Amount
Backplate		1
Outer Frame		1
Inner Frame		1
Gasket		1
Springs	T41071	1
Springs	T41320	1
Precision screws	F3ESN1	2
Precision screws bushing	F3ESN1P	2
Steel ball	Technische kogels RVS 420	1
Steel pins	1x10mm	4
Steel pins	3x10mm	4
Washers	3mm thick washer	4
Elliptical/circular mirror	BBE1-E02	1





1" mirrors comparison

	Ball-joint	Gimbal	Thor stand	Thor Gimbal
Resolution (°/rev)	0.47	0.47	0.46	1.00
Range (°)	±5	±5	±4	±15
Induced Translation (Dist: Surf – Cen (mm))	Yes (9.5)	No (0)	Yes (26.6)	No (0)
Dimension (mm)	53 × 53 × 22	70 × 57 × 23	50 x 50 x 25	54 x 54 x 28
Price (€)	32.17	32.00	40.71	316.58



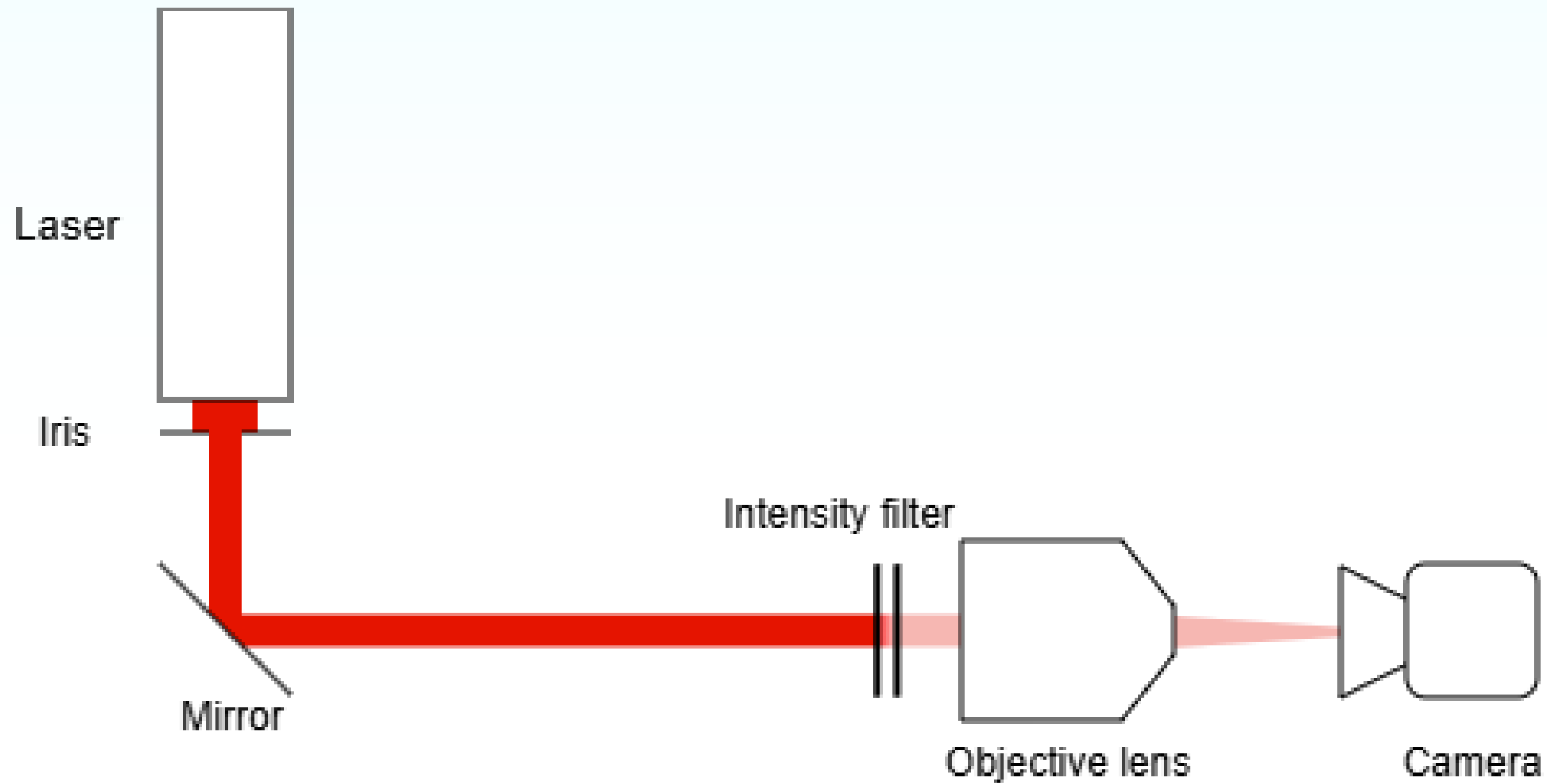


Elliptical mirrors comparison

	Ball-joint:	Gimbal:	Thorlabs 1":	Thorlabs 2":
Resolution (°/rev)	0.37	0.40	0.46	0.29
Range (°)	±5	±5	±4	±3
Induced Translation (Dist: Surf – Cen (mm))	Yes (9.5)	No (0)	Yes (43.9)	Yes (46.5)
Dimension (mm)	65 × 65 × 22	85 × 57 × 23	50 x 50 x 67	77 x 77 x 31
Price (€)	32.29	32.18	174.74	87.53

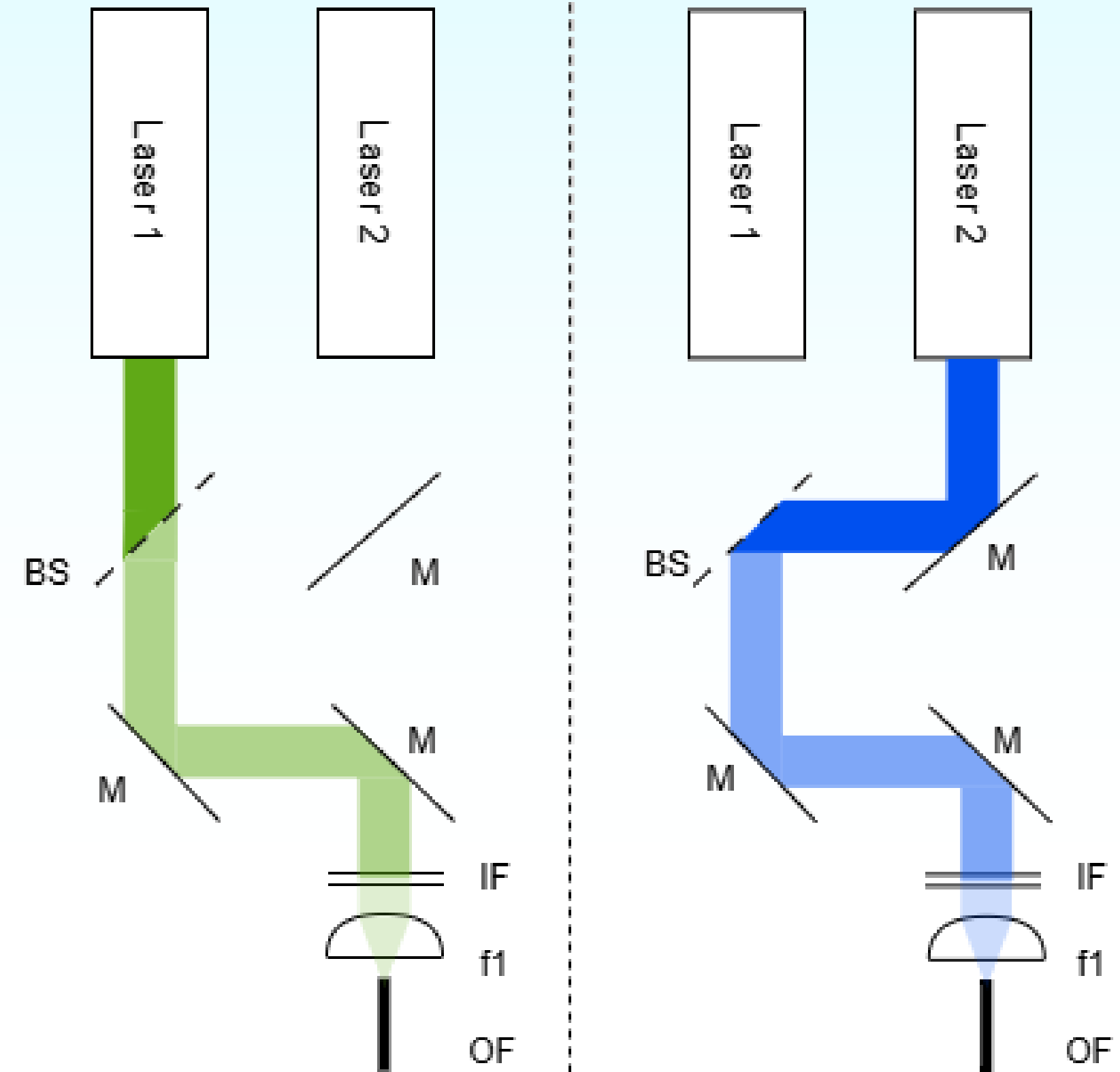


Mirror Drift Test



Laser module

- Extendable design
- Beamsplitter instead of dichroic



Legend: OF = optical fiber, M = mirror, BS = beamsplitter,
IF = intensity filter

f1 = 10 mm

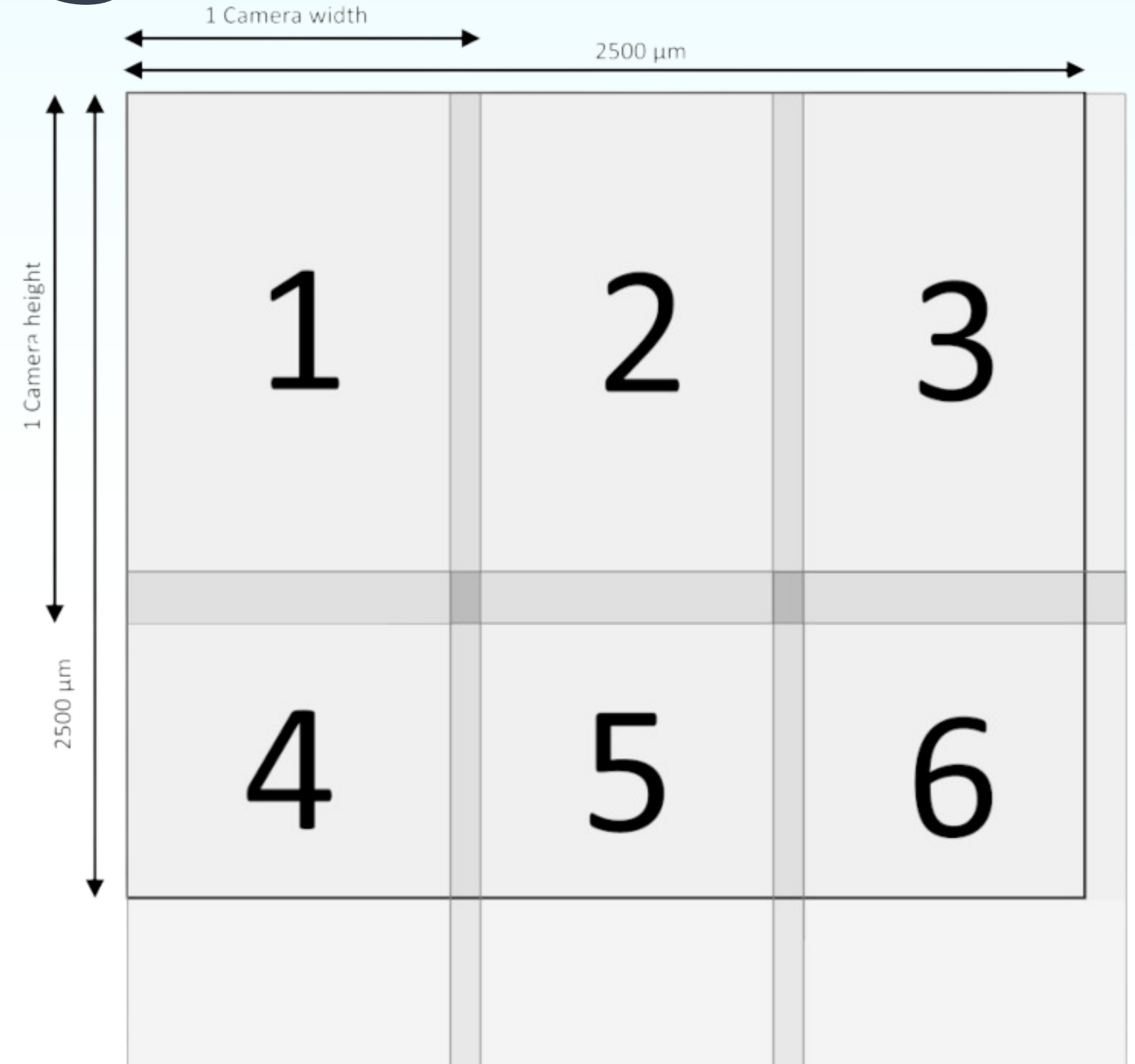




Image Stitching

- Stage vibrations
- Different Camera

FULL STAGE IN
5 min





1 —

Z-Stack

- Customizable steps
- Stage vibrations
- Different camera

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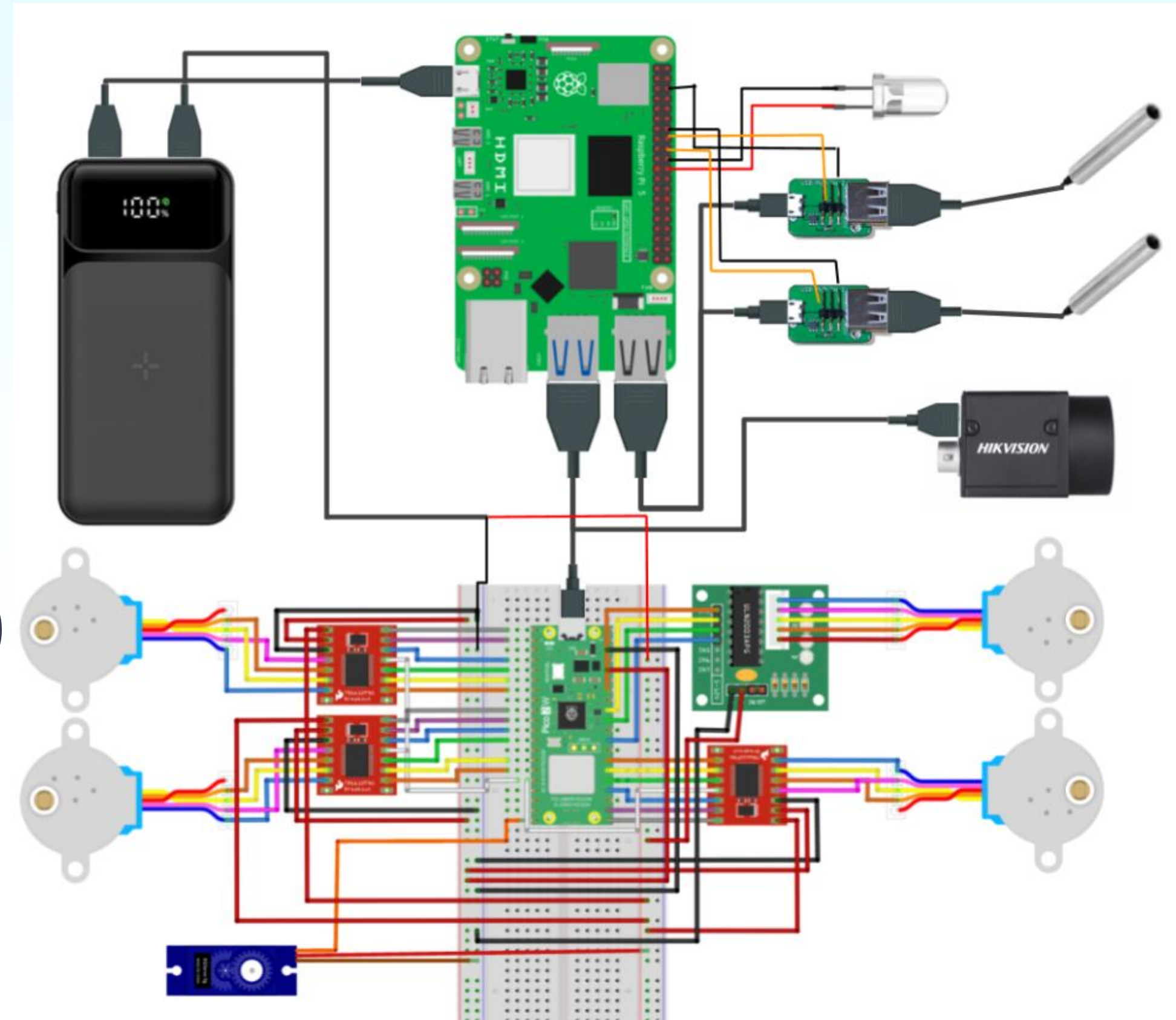
—

2 —



Electronics Layout

- Raspberry Pi 5 + Pico 2
- 4 stepper motors (28BYJ-48)
- 3 DC motor drivers (TB6612FNG)
- 1 stepper motor driver (ULN2003)
- Servomotor
- LED
- Camera
- Lasers + USB power switch





Interface

- User friendly
- Multiple usages



TURTLE Interface

Stage control

Individual axis

Axis: X axis

Steps: 1000

Delay us: 1500

Power %: 80

Move+

Move-

STOP

Release motors

Quick acces

X+

X-

Y+

Y-

Z+

Z-

Multi axis

X-steps: 1000

Y-steps: 1000

Delay us: 1500

Power %: 70

X+Y+

X+Y-

X-Y+

X-Y-

Position
X: 0.0 μ m
Y: 0.0 μ m
Z: 0.0 μ m

Home

Brightfield / Darkfield switch

Set Servo

Brightfield

Darkfield

Light source control

Laser 1 ON OFF

Laser 2 ON OFF

LED ON OFF

Filter array

Filter 1

Filter 2

Filter 3

Filter 4

Image capture

Format PNG

Filename

Picture 1

Capture

Start livestream

Use the toggle to overwrite livestream.jpg every 0.2 s. Capture still saves the selected image to disk.

Livestream preview

Image stitching

Set point 1

Reset point 1

Set point 2

Reset point 2

Set point 3

Reset point 3

Reset points/stitching

Point coordinates
Point 1: X=0.0 μ m, Y=0.0 μ m

StitchingImages

Start image stitch

Parameters

Objective magnification 4

Tube lens focal length (mm) 200

Imaging lens focal length (mm) 180

Camera resolution 3072x2048

Camera pixel grootte (μ m) 2.4

Camera FOV FOV X: 2048.0
FOV Y: 1365.3

Z-stack

Start z 0.0

Set current Z

End z 0.0

Set current Z

Steps 10

Set steps

Z-stack_Images

Start z-stack

Reset values

Console

```
{ "pico_response": { "ok": true }, "sent": { "angle": 0, "cmd": "servo" } } \n { "light": "led", "state": "ON" } \n { "current_filter": 3, "message": "Filter move started", "ok": true, "pico_response": { "sent": { "previous_filter": 1, "rotations": 2, "sent": { "axis": "filter", "cmd": "move", "delay_us": 5000, "power": 80, "steps": 4096 }, "steps": 4096, "target_filter": 3 } } \n { "current_filter": 1, "message": "Filter move started", "ok": true, "pico_response": { "sent": { "previous_filter": 3, "rotations": -2, "sent": { "axis": "filter", "cmd": "move", "delay_us": 5000, "power": 80, "steps": -4096 }, "steps": -4096, "target_filter": 1 } } \n Set stitching point 1: X=0.0, Y=0.0 \n
```



General Formulas

$$M = \frac{f_2}{f_1}$$

- M = magnification (-)
- f_1 = focal length of lens 1 (mm)
- f_2 = focal length of lens 2 (mm)

$$r = \frac{0.61\lambda}{NA}$$

- r = spatial resolution (μm)
- λ = wavelength (μm)
- NA = numerical aperture (-)

$$d_{pup} = 2 \cdot NA \cdot f_{obj}$$

- d_{pup} = diameter of the collimated beam out of the objective (mm)
- NA = numerical aperture (-)
- f_{obj} = focal length of the objective lens (mm)

$$L_{max} = \frac{f_{tube}(d_{tube} - d_{pup})}{d_{image\ sensor}}$$

- L_{max} = maximum length between the objective and tube lens (mm)
- f_{tube} = focal length of the tube lens (mm)
- d_{tube} = diameter of the tube lens (mm)
- d_{pup} = diameter of the collimated beam out of the objective (mm)
- $d_{image\ sensor}$ = diameter of the image sensor (mm)





PSF Formulas

$$FWHM(px) = 2\sqrt{2\ln(2)} d$$

- d = Standard deviation of the fitted Gaussian function (pixels)
- $FWHM(px)$ = Full Width at Half Maximum (pixels)

$$FWHM(\mu m) = FWHM(px) \times \frac{Ap}{M}$$

- $FWHM(\mu m)$ = Full Width at Half Maximum (μm)
- $FWHM(px)$ = Full Width at Half Maximum (pixels)
- M = Magnification (-)
- Ap = Pixel size of the camera (μm)





Estimated Magnification

$$\left. \begin{aligned} FOV_{per\ px} &= \frac{px_{size}}{M} \\ w_{line} &= FOV_{per\ px} \cdot \#px_{FWHM} \end{aligned} \right\} M = \frac{px_{size} \cdot \#px_{FWHM}}{w_{line}}$$

- M = magnification (-)
- px_{size} = pixel size (μm)
- $\#px_{FWHM}$ = number pixels for FWHM (-)
- w_{line} = width of the test line (μm)

Input variabels:

- $w_{line} = 7.8$ with $u_{w\ line} = 0.05$
- $px_{size} = 3.45$ with $u_{px\ size} = 0.005$
- $U_{px\ index} = 0.5$



Magnification estimate:
 $8,45 \pm 0,08$

Magnification vertical:
 $8,4 \pm 0,1$

Magnification horizontal:
 $8,5 \pm 0,1$





Uncertainty Formulas

The functional approach was used to identify the uncertainties during the result analysis:

$$f(x, y, z) = \dots$$

$$u_{f,x} = \left. \frac{\partial f}{\partial x} \right|_{y,z} \cdot u_x$$

$$u_{f,y} = \left. \frac{\partial f}{\partial y} \right|_{x,z} \cdot u_y$$

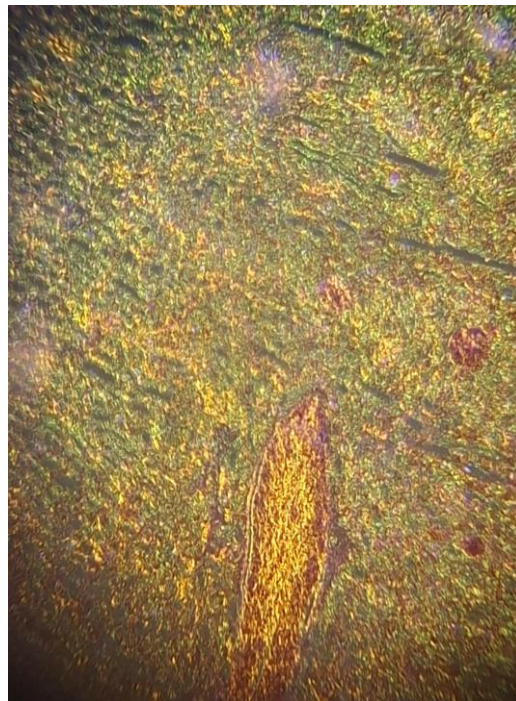
$$u_{f,z} = \left. \frac{\partial f}{\partial z} \right|_{x,y} \cdot u_z$$

$$u_f = \sqrt{u_{f,x}^2 + u_{f,y}^2 + u_{f,z}^2}$$





OpenUC2



- Open frame
- Very wobbly
- Adjusting with friction-hold flexures
- Somewhat modular





OpenFlexure

- Open frame
- Not modular



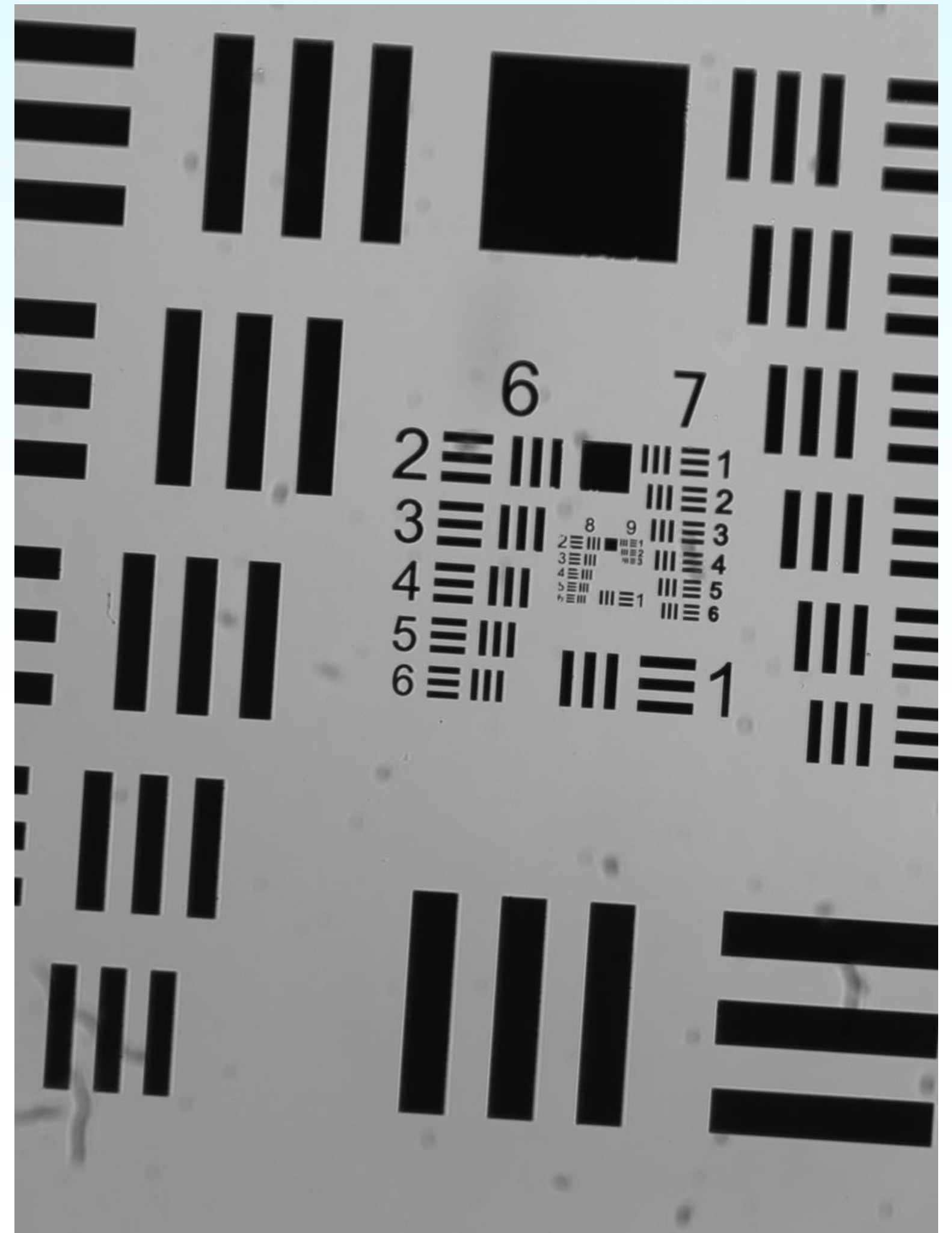


USAF PS75P

- No vignetting
- Even distribution of light

The interspacing width of the used elements:

- G6, E1 = 7,8 mm
- G8, E5 = 1,2 mm
- G8, E6 = 1,1 mm
- G9, E1 = 0,98 mm
- G9, E2 = 0,87 mm
- G9, E3 = 0,79 mm





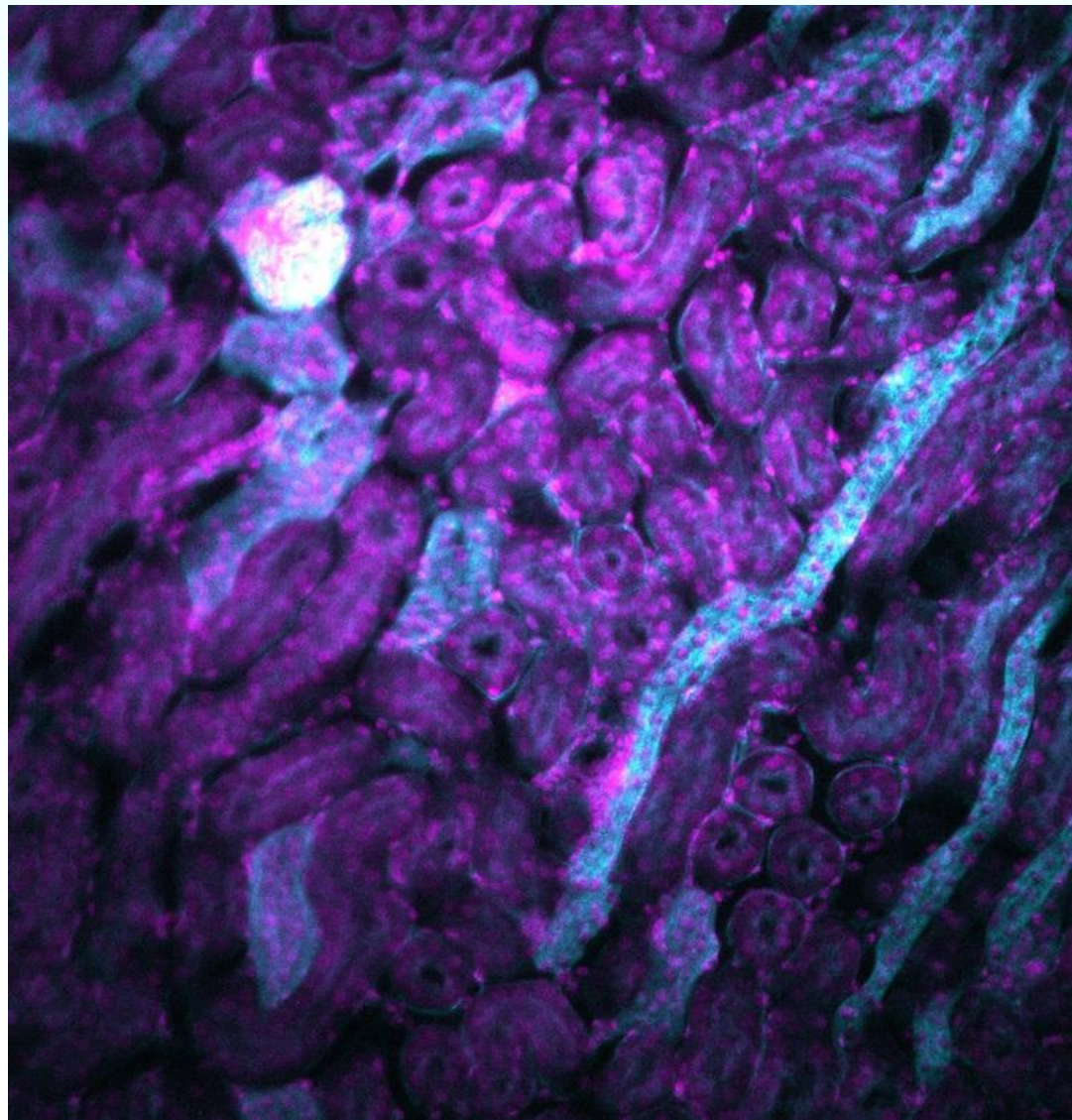
PSF Measurements

Bead	d in x (px)	FWHM in x (px)	FWHM in x (μm)	d in y (px)	FWHM in y (px)	FWHM in y (μm)
1	1.23634	2.9113582145	1.1830607585	1.57665	3.7127270240	1.5087053278
2	1.48051	3.4863346249	1.4167084165	1.25336	2.9514372516	1.1993472931
3	1.49474	3.5198437141	1.4303251842	1.49238	3.5142863388	1.4280668868
4	1.66154	3.9126276976	1.5899370503	1.97129	4.6420332066	1.8863385822
5	1.44359	3.3993946688	1.3813794591	1.35515	3.1911343840	1.2967507214
6	1.58481	3.7319423556	1.5165136781	1.30043	3.0622786312	1.2443888431
7	1.70539	4.0158865566	1.6318973640	1.20457	2.8365455816	1.1526598653
8	1.31586	3.0986135045	1.2591538976	1.38851	3.2696911807	1.3286730946
9	1.11429	2.6239524280	1.0662704213	1.63177	3.8425247049	1.5614499684
10	1.15824	2.7274467690	1.1083264255	1.65162	3.8892678828	1.5804445460
Average			1.36			1.42
Standard deviation			0.19748			0.22253
CV			14.53807			15.68552



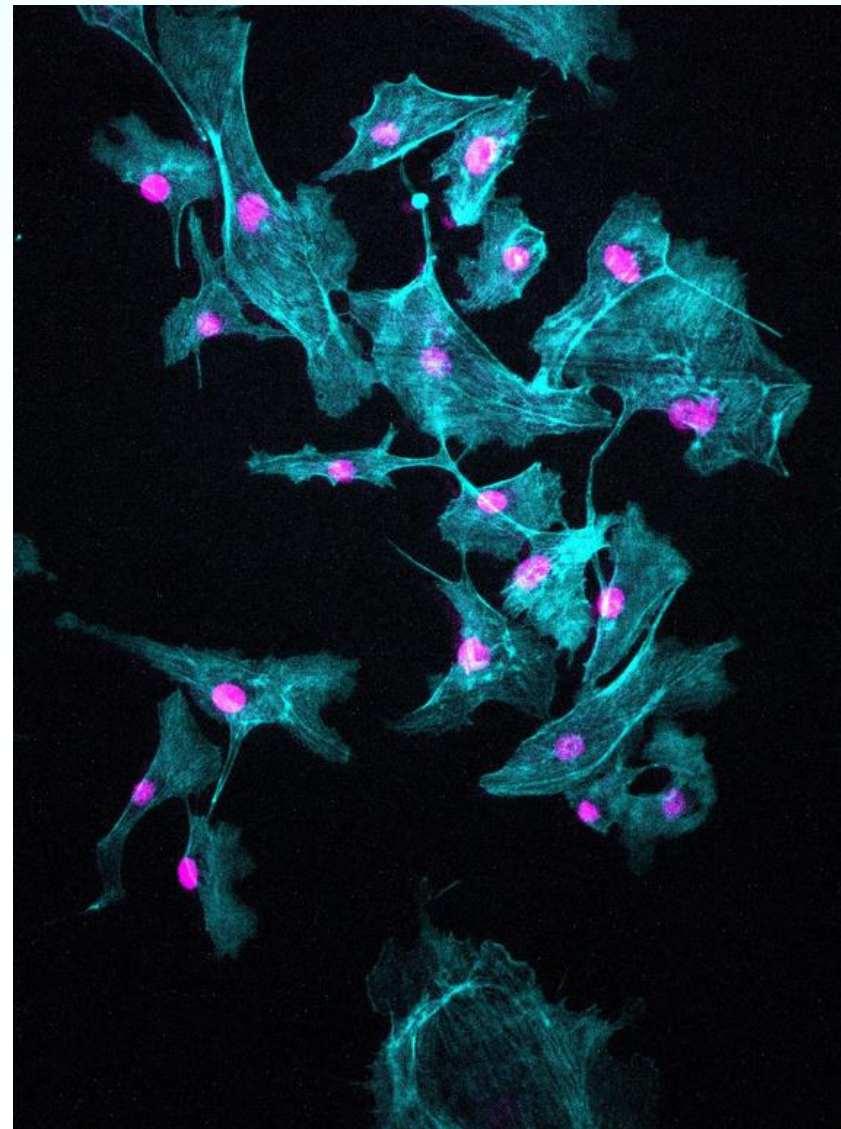


More images



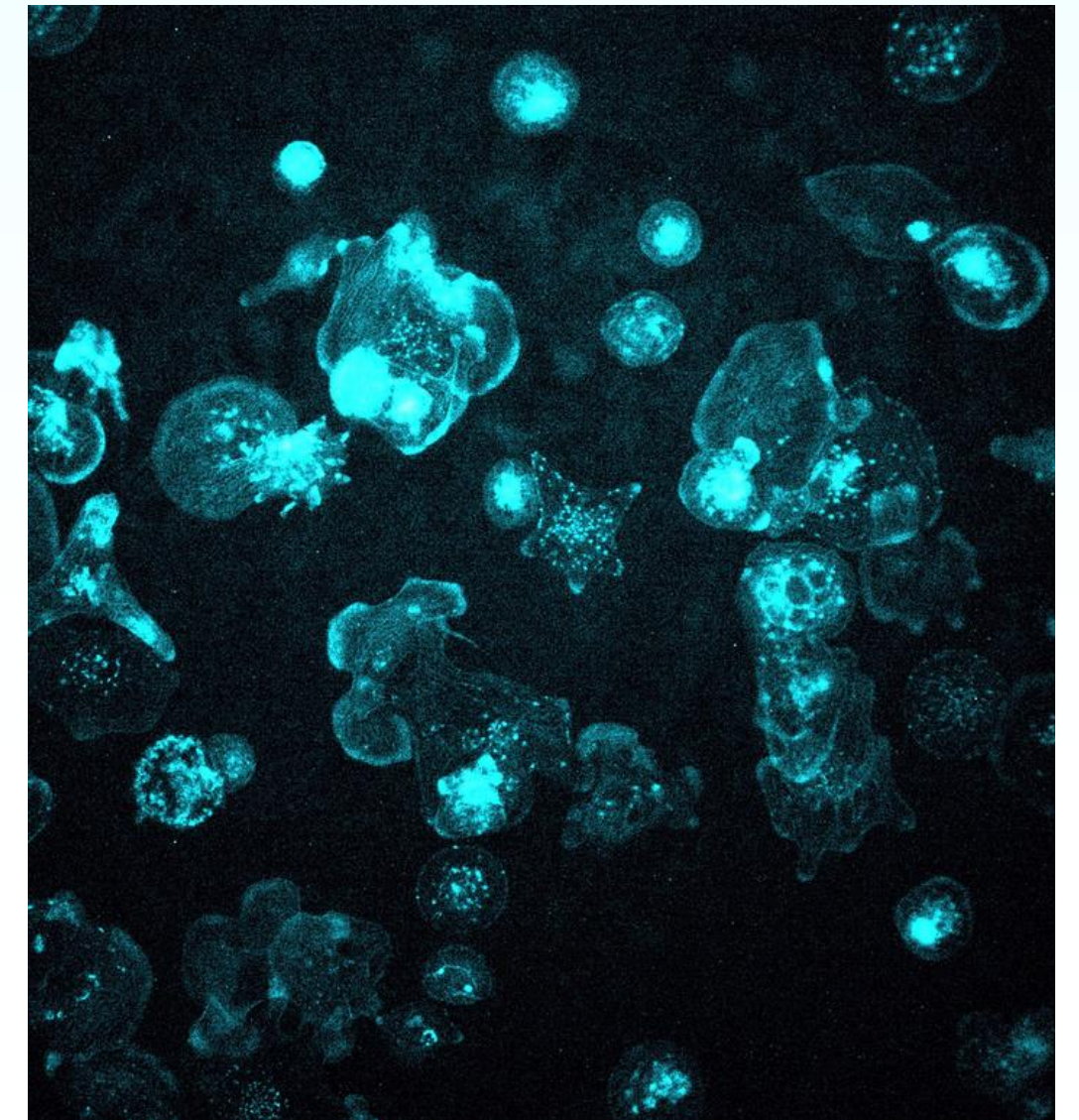
Mouse kidney

Thermo fischer FluoCells™



BPAAE cells

Thermo fischer FluoCells™



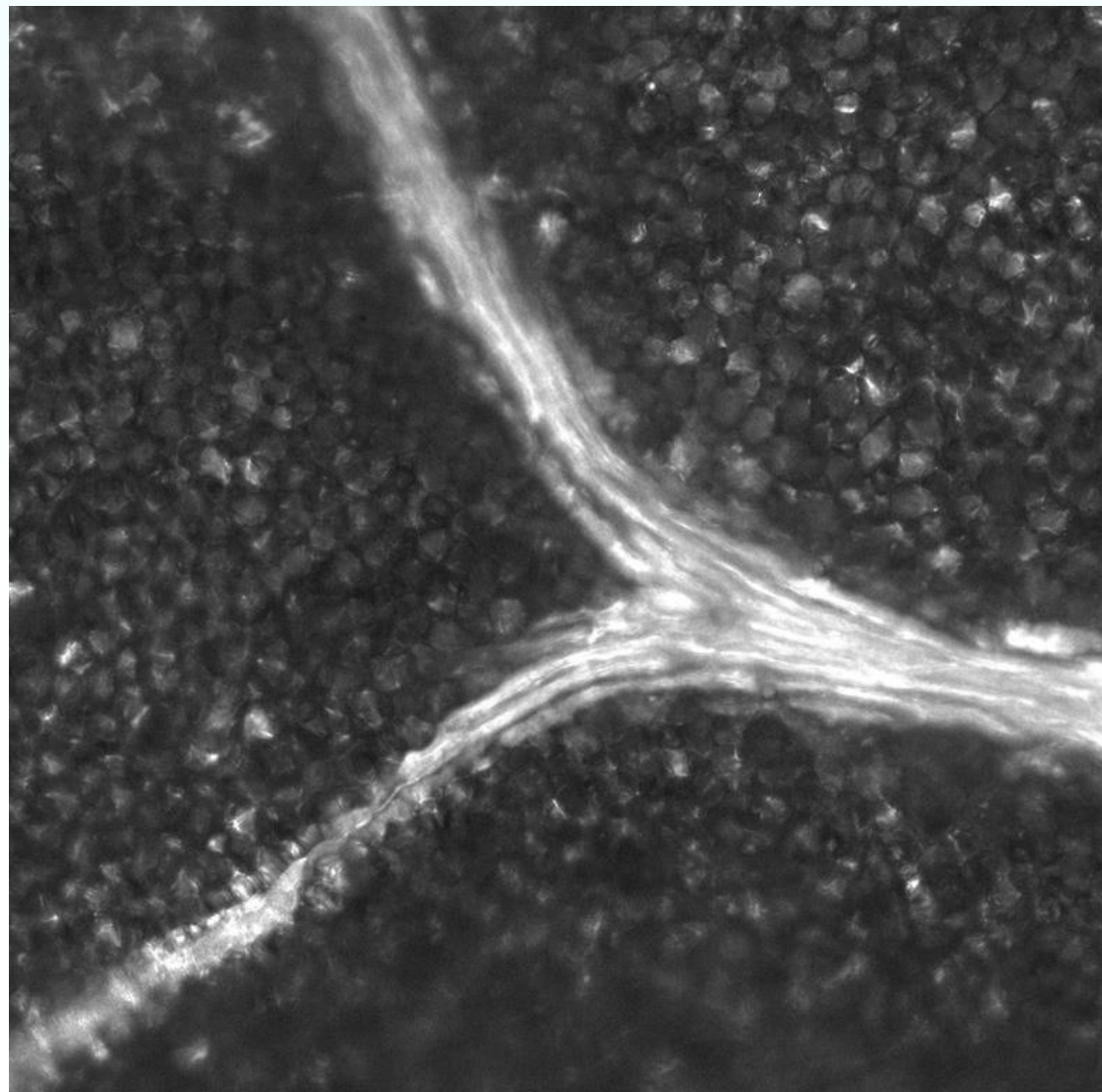
Macrophages

Sample supplied by
Roderick

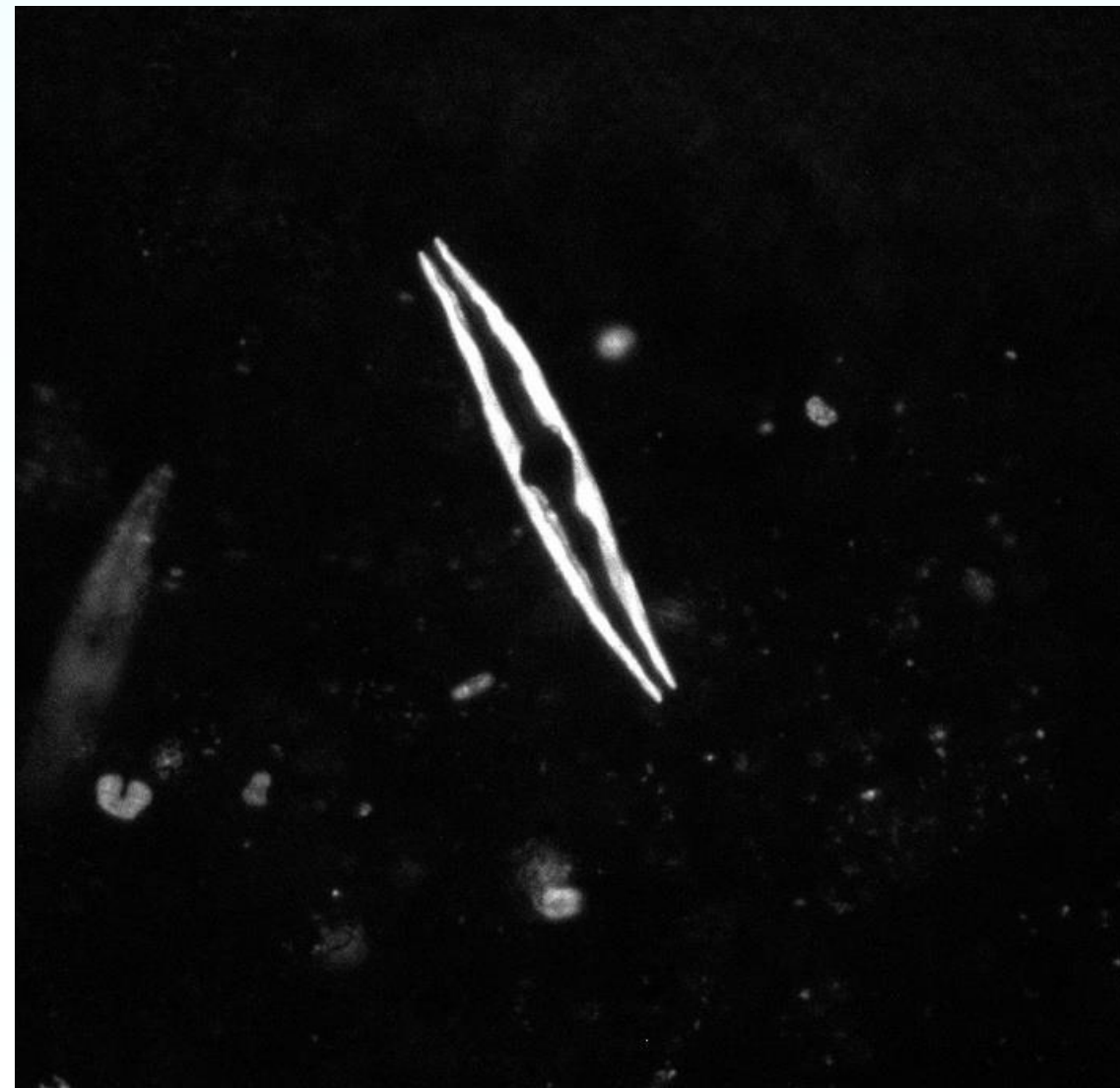




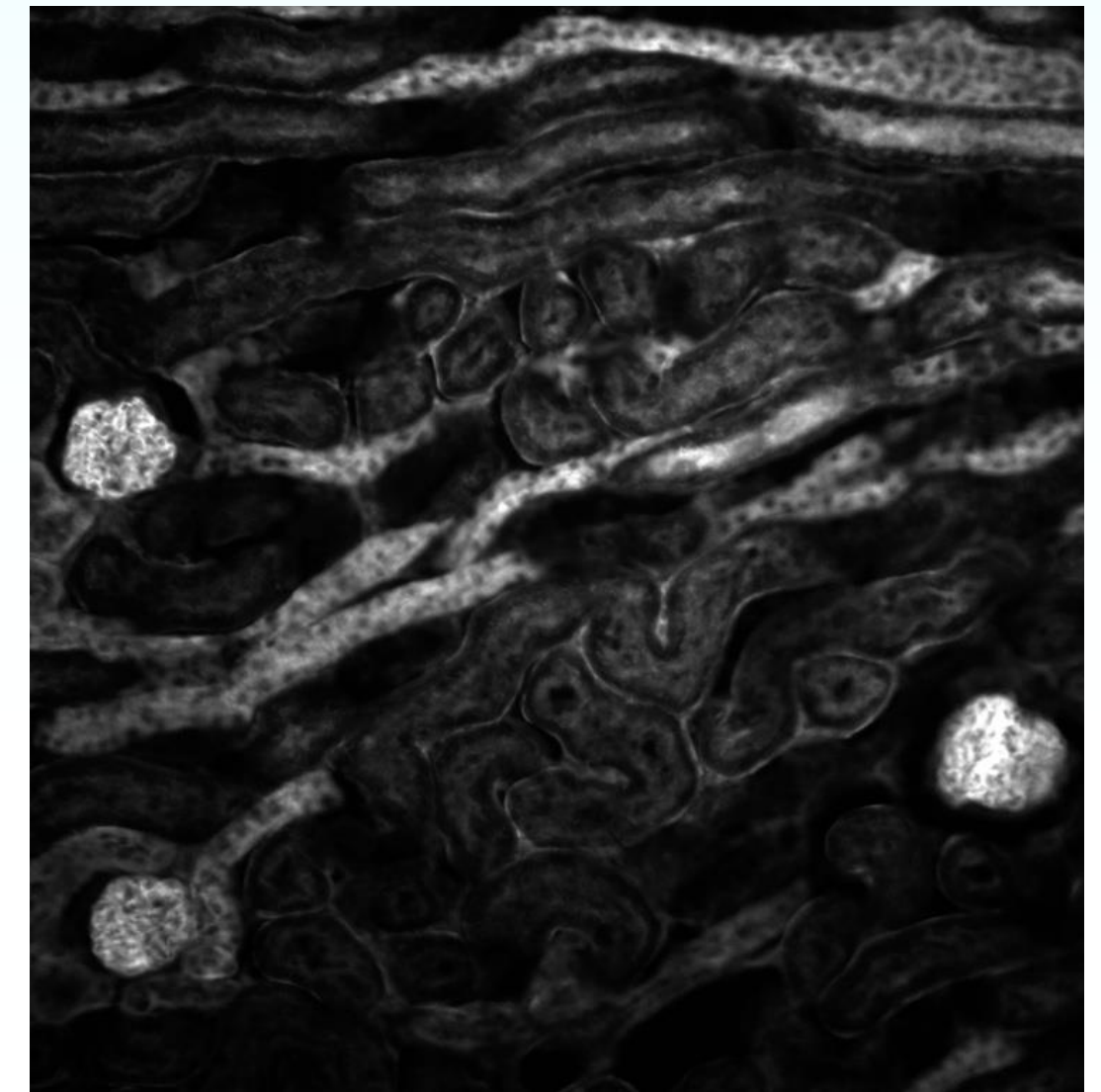
More images



Leaf



Diatom (living sample)



Mouse kidney

Thermo fischer FluoCells™





Stage

